arXiv:2308.01818v1 [math.FA] 3 Aug 2023

DUALITY, BMO AND HANKEL OPERATORS ON BERNSTEIN SPACES

CARLO BELLAVITA, MARCO M. PELOSO

Abstract. In this paper we deal with the problem of describing the dual space $(\mathcal{B}^1_\kappa)^*$ of the Bernstein space $\mathcal{B}^1_\kappa$, that is the space of entire functions of exponential type (at most) $\kappa > 0$ whose restriction to the real line is Lebesgue integrable. We provide several characterizations, showing that such dual space can be described as a quotient of the space of entire functions of exponential type κ whose restriction to the real line are in a suitable BMO-type space, or as the space of symbols b for which the Hankel opertor H_b is bounded on the Paley–Wiener space $\mathcal{B}^2_{\kappa/2}$. We also provide a characterization of $(\mathcal{B}^1_\kappa)^*$ as the BMO space w.r.t. the Clark measures of the inner function $e^{i\kappa z}$ on the upper half-plane, in analogy with the known description of the dual of backward-shift invariant 1-spaces on the torus. Furthermore, we show that the orthogonal projection $P_\kappa : L^2(\mathbb{R}) \to \mathcal{B}^2_\kappa$ induces a bounded operator form $L^\infty(\mathbb{R})$ onto $(\mathcal{B}^1_\kappa)^*$.

Finally, we show that $\mathcal{B}^1_\kappa$ is the dual space of the suitable VMO-type space or as the space of symbols b for which the Hankel opertor H_b on the Paley–Wiener space $\mathcal{B}^2_{\kappa/2}$ is compact.

1. Introduction and statement of the main results

Let $\mathcal{E}_\kappa$ be the space of entire functions of exponential type at most κ,

$$\mathcal{E}_\kappa = \left\{ f \in \mathrm{Hol}(\mathbb{C}) : \text{ for every } \varepsilon > 0 \text{ there exists } C_\varepsilon > 0 \text{ such that } \ |f(z)| \leqslant C_\varepsilon e^{(\kappa+\varepsilon)|z|} \right\}. \quad (1)$$

For $p \in (0, \infty]$, the Bernstein space $\mathcal{B}^p_\kappa$ is defined as

$$\mathcal{B}^p_\kappa = \left\{ f \in \mathcal{E}_\kappa : f_0 \in L^p,\ \|f\|_{\mathcal{B}^p_\kappa} = \|f_0\|_{L^p} \right\}$$

where $f_0 := f|_{\mathbb{R}}$ denotes the restriction of f to the real line and L^p is the standard Lebesgue space. When $p = 2$, the space $\mathcal{B}^2_\kappa$ is the classical Paley–Wiener space PW_κ. We remark that the spaces $\mathcal{B}^p_\kappa$ are often also called Paley–Wiener spaces and denoted as PW^p_κ. We select to refer them as Bernstein spaces, see the discussion at the end of this introduction.

Let $\mathcal{S}$ and $\mathcal{S}'$ denote the space of Schwartz functions and the space of tempered distributions on the real line $\mathbb{R}$, resp. For $f \in \mathcal{S}'$ we equivalently denote by $\widehat{f}$ or $\mathcal{F}f$ the Fourier transform, that for $f \in \mathcal{S}$ it is given by

$$\widehat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{-ix\xi}\, dx.$$

The Fourier transform $\mathcal{F}$ is an isomorphism of $\mathcal{S}$ onto itself with inverse given by

$$\mathcal{F}^{-1} g(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(\xi) e^{ix\xi}\, d\xi,$$

and by Plancherel Theorem, $\mathcal{F}$ extends to a surjective isometry $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$.

2000 *Mathematics Subject Classification.* 30D15, 30H35, 47B35.

Key words and phrases. Bernstein spaces, Paley–Wiener spaces, BMO, duality, Hankel operators.

M. Peloso is member of the Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni (GNAMPA) of the Istituto Nazionale di Alta Matematica (INdAM). He was partially supported by the 2022 INdAM–GNAMPA project *Holomorphic Functions in One and Several Complex Variables* (CUP_E55F22000270001).

R. Paley and N. Wiener [PW34] showed that $\mathcal{B}^2_\kappa$ can be characterized as the subspace of L^2 consisting of the functions whose Fourier transform is supported in $[-\kappa,\kappa]$. There exists an analogous characterization of the Bernstein space $\mathcal{B}^p_\kappa$ as the subspace of L^p consisting of the functions whose Fourier transforms, as tempered distributions, are supported in $[-\kappa,\kappa]$, see e.g. [And14]. The Paley–Wiener space $\mathcal{B}^2_\kappa$ is a reproducing kernel Hilbert space with reproducing kernel K_w given by a dilated of the normalized sinc-function, namely

$$K_w(z) = \sqrt{\tfrac{\kappa}{\pi}}\operatorname{sinc}\left(\tfrac{\kappa}{\pi}(z-\overline{w})\right) =: K(z,w),$$

where we write $\operatorname{sinc}(z) = \frac{\sin(\pi z)}{\pi z}$. The set $\left\{\frac{1}{\sqrt{2\kappa}}e^{-i\frac{\pi}{\kappa}n(\cdot)} : n\in\mathbb{Z}\right\}$ is an orthonormal basis of $L^2[-\kappa,\kappa]$, so that

$$\left\{\sqrt{\tfrac{\kappa}{\pi}}\operatorname{sinc}\left(\tfrac{\kappa}{\pi}(\cdot-\tfrac{\pi}{\kappa}n)\right) : n\in\mathbb{Z}\right\} = \left\{K_{\frac{\pi}{\kappa}n} : n\in\mathbb{Z}\right\}$$

is an orthonormal basis of $\mathcal{B}^2_\kappa$. Notice in particular that this is an orthonormal basis of reproducing kernels. In particular, for $f\in\mathcal{B}^2_\kappa$ we have the reproducing formula[1]

$$f(z) = \sum_{n\in\mathbb{Z}}\langle f \,|\, K_{\frac{\pi}{\kappa}n}\rangle K_{\frac{\pi}{\kappa}n}(z) = \sum_{n\in\mathbb{Z}} f\left(n\tfrac{\pi}{\kappa}\right)\operatorname{sinc}\left(\tfrac{\kappa}{\pi}(z-\tfrac{\pi}{\kappa}n)\right), \tag{2}$$

and the renowed Whittaker–Kotelnikov–Shannon formula

$$\|f\|^2_{\mathcal{B}^2_\kappa} = \sum_{n\in\mathbb{Z}}\left|f\left(\tfrac{\pi}{\kappa}n\right)\right|^2. \tag{3}$$

When $p\neq 2$, analouges of formulas (2) and (3) hold. Precisely, if $p\in(1,\infty)$ and $a=(a_n)\in\ell^p(\mathbb{Z})$, then

$$f(z) := \sum_{n\in\mathbb{Z}} a_n\sqrt{\tfrac{\kappa}{\pi}}\operatorname{sinc}\left(\tfrac{\kappa}{\pi}z-n\right) \tag{4}$$

belongs to $\mathcal{B}^p_\kappa$ and moreover $f(\frac{\kappa}{\pi}n) = a_n$, for $n\in\mathbb{Z}$. The convergence of the above series is in $\mathcal{B}^p_\kappa$-norm and therefore also uniformly on compact subsets of $\mathbb{C}$. Conversely, if $f\in\mathcal{B}^p_\kappa$, then $a := \left(f\left(\frac{\kappa}{\pi}n\right)\right)_{n\in\mathbb{Z}}\in\ell^p(\mathbb{Z})$ and

$$f(z) = \sum_{n\in\mathbb{Z}} f(\frac{\kappa}{\pi}n)\sqrt{\tfrac{\kappa}{\pi}}\operatorname{sinc}\left(\tfrac{\kappa}{\pi}z-n\right).$$

Moreover, $\|f\|_{\mathcal{B}^p_\kappa}\asymp\|a\|_{\ell^p(\mathbb{Z})}$, [PP37]. Hence, when $p\in(1,\infty)$, the correspondence $\mathcal{B}^p_\kappa\ni f\mapsto\left(f(\frac{\kappa}{\pi}n)\right)\in\ell^p$ is an isomorphism of $\mathcal{B}^p_\kappa$ onto ℓ^p. We also refer to [PW34] and [Lev96, Lectures 20-21] for more on the spaces $\mathcal{B}^p_\kappa$, with $p\in[1,\infty]$.

Let $P_\kappa : L^2\to\mathcal{B}^2_\kappa$ be the orthogonal projection. Then, for $f\in L^2$,

$$P_\kappa f(x) = f * \sqrt{\tfrac{\kappa}{\pi}}\operatorname{sinc}\left(\tfrac{\kappa}{\pi}\,\cdot\,\right)(x).$$

From the boundedness of the Hilbert trasform on L^p, $p\in(1,\infty)$, it follows at once that the dual of $\mathcal{B}^p_\kappa$ can be identified with $\mathcal{B}^{p'}_\kappa$, under the standard (bilinear) pairing of duality

$$L_g : \mathcal{B}^p_\kappa\ni f\mapsto\int_{\mathbb{R}} f g\,dm,$$

for $g\in\mathcal{B}^{p'}_\kappa$, $p\in(1,\infty)$, $\frac{1}{p}+\frac{1}{p'}=1$. It is worth pointing out that the spaces $\mathcal{B}^p_\kappa$ are endowed with an involution $\mathcal{B}^p_\kappa\ni f\mapsto f^\#\in\mathcal{B}^p_\kappa$, where $f^\#(z) := \overline{f(\overline{z})}$. Thus, the above pairing of duality can

[1] We denote by $\langle\cdot\,|\,\cdot\rangle$ a sesquilinear pairing, and by $\langle\cdot,\cdot\rangle$ a bilinear pairing.

be equivalently replaced by the sesquilinear pairing $\langle f \,|\, g\rangle = \int_{\mathbb{R}} f\overline{g}\, dm$. Moreover, by (4), for $f \in \mathcal{B}^p_\kappa$, $g \in \mathcal{B}^{p'}_\kappa$,

$$\int_{\mathbb{R}} f(x)g(x)\, dx = \sum_{n\in\mathbb{Z}} f\big(\tfrac{\kappa}{\pi}n\big)g\big(\tfrac{\kappa}{\pi}n\big).$$

In [Eof95], C. Eoff extended the above isomorphisms between $\mathcal{B}^p_\kappa$ and ℓ^p to the cases $p \in (0,1]$ – however in this discussion, we limit ourselves to the case $p = 1$. Let $H_{d,\alpha}$ denote the α-translated discrete Hilbert transform[2]

$$H_{d,\alpha}a(n) = \sum_{k\neq n} \frac{a_k}{n-k+\alpha}, \tag{5}$$

where $a = (a_n) \in \ell^1$. Then, we define $H^1 = H^1(\mathbb{Z})$ the Hardy space on $\mathbb{Z}$ as

$$H^1 = \big\{a = (a_n)_{n\in\mathbb{Z}} :\ a,\ H_{d,\alpha}a \in \ell^1\big\},$$

for one (equivalently, all) $\alpha \in [0,1)$, with norm $\|a\|_{H^1} = \|a\|_{\ell^1} + \|H_{d,\alpha}a\|_{\ell^1}$. Eoff proved that if $f \in \mathcal{B}^1_\pi$ then $a := \big((-1)^n f\big(\frac{\kappa}{\pi}n\big)\big)_{n\in\mathbb{Z}} \in H^1$, and conversely, it $a = (a_n) \in H^1$, there exists a unique $f \in \mathcal{B}^1_\pi$ such that $(-1)^n f\big(\frac{\kappa}{\pi}n\big) = a_n$, for all $n \in \mathbb{Z}$. Moreover, $\|f\|_{\mathcal{B}^1_\pi} \asymp \big\|\big((-1)^n f\big(\frac{\kappa}{\pi}\cdot\big)\big)\big\|_{H^1}$. The dual of $H^1(\mathbb{Z})$ can be identified with $\mathrm{BMO}(\mathbb{Z})$ with the pairing of duality

$$\langle h, b\rangle := \sum_{n\in\mathbb{Z}} h_n b_n,$$

where $h = (h_n) \in H^1(\mathbb{Z})$, $b = (b_b) \in \mathrm{BMO}(\mathbb{Z})$. The above pairing of duality must be interpreted in a weak sense, since the sum may not converge absolutely, in general, using the atomic characterization of $H^1(\mathbb{Z})$ proved in [BC98]. Moreover, the $\mathrm{BMO}(\mathbb{Z})$-norm is equivalent to the norm of duality with $H^1(\mathbb{Z})$.

Given the results cited above, it is fairly straightforward to obtain a description of the dual of $\mathcal{B}^1_\kappa$ as a space of sequences. Precisely, given any $L \in (\mathcal{B}^1_\kappa)^*$, there exists a unique $b \in \mathrm{BMO}(\mathbb{Z})$ such that

$$L(f) = \big\langle\big((-1)^n f(n)\big), b\big\rangle,$$

with $\|L\|_{(\mathcal{B}^1_\kappa)^*} \asymp \|b\|_{\mathrm{BMO}(\mathbb{Z})}$, and conversely any $b \in \mathrm{BMO}(\mathbb{Z})$ defines a continuous linear functional on $H^1(\mathbb{Z})$. Such a description of the dual space $(\mathcal{B}^1_\kappa)^*$ is analogous with the characterization of the dual space of coinvariant subspaces of $H^1(\mathbb{D})$ [Bes15], where $\mathbb{D}$ denotes the unit disk, and $H^1(\mathbb{D})$ is the classical Hardy space on $\mathbb{D}$; see also [O'L22].

The main goal of this work is to describe the dual of $\mathcal{B}^1_\kappa$ as a space of entire functions, extending the pairing of duality between $\mathcal{B}^p_\kappa$ and $\mathcal{B}^{p'}_\kappa$, $p \in (1,\infty)$, $\frac{1}{p}+\frac{1}{p'} = 1$. One of characterizations that we obtain involves, in a rather natural and classical way, the symbols of bounded Hankel operators, that we now describe. Given a locally integrable function φ on $\mathbb{R}$, let H_φ denote the Hankel operator on $\mathcal{B}^2_\kappa$ (that is, the Paley–Wiener space PW_κ) with symbol φ, given by,

$$H_\varphi(f) = P_\kappa(\varphi\overline{f}) : \mathcal{B}^2_\kappa \to \mathcal{B}^2_\kappa, \tag{6}$$

initially defined for f in a dense subspace of $\mathcal{B}^2_\kappa$. Such operators where first studied by R. Rochberg [Roc87] who obtained a characterization of bounded and compact operators, in terms of properties of their symbols. We refer to Section 3 for a discussion of these results.

Before stating our main results, we need a few more definitions and remarks. Since, for $\lambda > 0$, the mapping $\mathcal{B}^p_\kappa \ni f \mapsto \lambda^{-1/p} f(\lambda\cdot) \in \mathcal{B}^p_{\lambda\kappa}$ is an isometric isomorphism, without loss of generality,

[2]The restriction to summing on $\mathbb{Z}\backslash\{n\}$ is required only when $\alpha = 0$.

we may take $\kappa = \pi$. Since there is no loss of generality in considering only the case $\kappa = \pi$, we are going to do so throught the paper.

Let $\mathrm{BMO}(\mathbb{R})$ denote the space of functions (modulo constants) of bounded mean oscillation:

$$\mathrm{BMO}(\mathbb{R}) = \Big\{ \varphi \in L^1_{\mathrm{loc}}(\mathbb{R}) : \|\varphi\|_{\mathrm{BMO}(\mathbb{R})} := \sup_{I \subset \mathbb{R}} \frac{1}{|I|} \int_I |\varphi - \varphi_I| \, dm < \infty \Big\},$$

where $I \subset \mathbb{R}$ is any bounded interval, $\varphi_I = \frac{1}{|I|}\int_I \varphi$ is the average of φ over I, and $|E|$ denotes the Lebesgue measure of a measurable set E.[3] We also define the space of analytic BMO,

$$\mathrm{BMOA} = \big\{ \varphi \in \mathrm{BMO}(\mathbb{R}) : \mathrm{supp}(\widehat{\varphi}) \subseteq (0, \infty) \big\}.$$

Notice that $\overline{\mathrm{BMOA}} = \big\{ \varphi \in \mathrm{BMO}(\mathbb{R}) : \mathrm{supp}(\widehat{\varphi}) \subseteq (-\infty, 0) \big\}$.

We define a space of equivalence classes of functions in $\mathcal{E}_\pi$.

Definition 1.1. Let

$$\mathrm{BMO}(e^{-i\pi z}) = \Big\{ f \in \mathcal{E}_\pi : e^{i\pi(\cdot)} f_0 \in \mathrm{BMOA}, \ e^{-i\pi(\cdot)} f_0 \in \overline{\mathrm{BMOA}} \Big\} \Big/ \mathrm{span}\{ e^{\pm i\pi(\cdot)} \Big\},$$

with norm

$$\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = \inf_{c \in \mathbb{C}} \|e^{i\pi(\cdot)} f_0 + c e^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} + \inf_{c \in \mathbb{C}} \|e^{-i\pi(\cdot)} f_0 + c e^{-i2\pi(\cdot)}\|_{\overline{\mathrm{BMOA}}}. \tag{7}$$

Remarks 1.2.

(1) Observe that the definition of $\mathrm{BMO}(e^{-i\pi z})$ makes sense, since the functions $g_\pm := e^{\pm i\pi z} \in \mathcal{E}_\pi$ and that $e^{\pm i\pi z} g_\pm \in L^\infty \hookrightarrow \mathrm{BMO}(\mathbb{R})$.

(2) Notice that if $f \in \mathrm{BMO}(e^{-i\pi z})$, then $f_0 \in \mathcal{S}'$, and by the well-known properties of $\mathrm{BMO}(\mathbb{R})$, $\int_{\mathbb{R}} |f(x)| \frac{dx}{1+x^2} < \infty$.

(3) With the given norm, $\mathrm{BMO}(e^{-i\pi z})$ is a Banach space, see Theorem 2.3.

Our first result is the identification of $(\mathcal{B}^1_\pi)^*$ with the spaces $\mathrm{BMO}(e^{-i\pi z})$. Recall that,

$$(\mathcal{B}^1_\pi)_{|_{\mathbb{R}}} = \big\{ g \in L^1(\mathbb{R}) : \mathrm{supp}(\widehat{g}) \subseteq [-\pi, \pi] \big\},$$

see e.g. [And14, CP22a]. This implies that, since if $f \in \mathcal{B}^1_\pi$, $\widehat{f}_0(\pm\pi) = 0$, the bounded functions $e^{\pm i\pi(\cdot)}$ annihilate $\mathcal{B}^1_\pi$. Moreover, we point out the subspace $\{ f \in \mathcal{B}^1_\pi : f_0 \in \mathcal{S} \}$ is dense in $\mathcal{B}^1_\pi$, see Lemma 2.1.

Theorem 1. (1) *The projection P_π induces a surjective, bounded linear operator $P_\pi : L^\infty \to \mathrm{BMO}(e^{-i\pi z})$ such that $\|f\|_{\mathrm{BMO}(e^{-i\pi z})} \approx \inf\{\|\varphi\|_{L^\infty} : P_\pi \varphi = f\}$.*

(2) *The space $\mathrm{BMO}(e^{-i\pi z})$ can be identified with $(\mathcal{B}^1_\pi)^*$. Precisely, each $f \in \mathrm{BMO}(e^{-i\pi z})$ defines a bounded linear functional on $\mathcal{B}^1_\pi$ by setting*

$$L_f(h) = \int_{\mathbb{R}} h(x) f(x) \, dx$$

for any $h \in \mathcal{B}^1_\pi$, and conversely, for each $L \in (\mathcal{B}^1_\pi)^*$ there exists a unique $f \in \mathrm{BMO}(e^{-i\pi z})$ such that $L = L_f$. Moreover, $\|L_f\|_{(\mathcal{B}^1_\pi)^*} \approx \|f\|_{\mathrm{BMO}(e^{-i\pi z})}$.

[3] Adopting the same notation for $\mathrm{BMO}(\mathbb{R})$ and $\mathrm{BMO}(\mathbb{Z})$ should cause no confusion.

The above pairings of duality are not absolutely convergent in general and must be interpreted in a weak sense. Given $h \in \mathcal{B}^1_\pi$, let $h_{(\delta)} \in \mathcal{S}_\pi$ and $h_{(\delta)} \to h$ in $\mathcal{B}^1_\pi$ as $\delta \to 0^+$, then

$$\int_{\mathbb{R}} h(x) f(x)\, dx := \lim_{\delta \to 0^+} \int_{\mathbb{R}} h_{(\delta)}(x) f(x)\, dx; \tag{8}$$

the existence of such functions $h_{(\delta)}$ will be guaranteed by Lemma 2.1.

We now characterize the dual space $(\mathcal{B}^1_\pi)^*$ as the image of $L^\infty(\mathbb{R})$ via the orthogonal projection P_π and in terms of bounded Hankel operators on $\mathcal{B}^2_{\pi/2}$.

Theorem 2. *Let $\varphi \in \mathcal{S}'$ be a locally integrable function. Then, the following conditions are equivalent:*

(1) *there exists $\widetilde{\varphi} \in L^\infty$ such that $P_\pi(\widetilde{\varphi}) = \varphi$;*
(2) *φ defines an element of $(\mathcal{B}^1_\pi)^*$ with the pairing of duality as in (8);*
(3) *the Hankel operator with symbol φ, $H_\varphi : \mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}$ is bounded.*

Moreover, the quantities

$$\inf_{\widetilde{\varphi} \in L^\infty :\, P_\pi(\widetilde{\varphi}) = \varphi} \|\widetilde{\varphi}\|_{L^\infty}, \quad \|\varphi\|_{(\mathcal{B}^1_\pi)^*}, \quad \|H_\varphi\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}}$$

are all comparable.

As metioned earlier, it has been *folklore* for sometime that the dual space $(\mathcal{B}^1_\pi)^*$ can be described as a space of sequences. We now make this description explicit.

Definition 1.3. Let $\alpha \in [0, 1)$. We define

$$\mathcal{X}_\alpha = \Big\{ f \in \mathcal{E}_\pi :\, f(iy) = o(|y| e^{\pi |y|}) \text{ for } |y| \to \infty,\ \big(e^{i\pi(\cdot + \alpha)} f(\cdot + \alpha)\big)_{|\mathbb{R}} \in \mathrm{BMO}(\mathbb{Z}) \Big\} \Big/ \operatorname{span}\{e^{\pm i\pi(\cdot)}\},$$

with norm

$$\|f\|_{\mathcal{X}_\alpha} = \inf_{c_\pm \in \mathbb{C}} \big\| e^{i\pi(\cdot + \alpha)} \big(f(\cdot + \alpha) + c_+ e^{i\pi(\cdot)} + c_- e^{-i\pi(\cdot)} \big) \big\|_{\mathrm{BMO}(\mathbb{Z})} = \big\| e^{i\pi(\cdot + \alpha)} \big(f(\cdot + \alpha) \big\|_{\mathrm{BMO}(\mathbb{Z})}.$$

Remarks 1.4.

(1) Observe that the definition of $\mathcal{X}_\alpha$ makes sense, since the functions $e^{\pm i\pi z}$ are in $\mathcal{E}_\pi$ and satisfy the two other conditions in the definition of $\mathcal{X}_\alpha$, namely that $f(iy) = o(|y| e^{\pi |y|})$ for $|y| \to \infty$, and $\big(e^{i\pi(\cdot + \alpha)} f(\cdot + \alpha)\big)_{|\mathbb{Z}} \in \mathrm{BMO}(\mathbb{Z})$. Moreover, notice that $\mathcal{X}_\alpha \neq \{0\}$ since it contains $\operatorname{sinc}(\cdot - \alpha)$.

(2) With the given norm, $\mathcal{X}_\alpha$ is a Banach space, see Proposition 4.1.

(3) We point out that the spaces $\mathcal{X}_\alpha$ are related to the BMO spaces with respect to the the Clark measures of the inner function $e^{i2\pi z}$, see the remark in Section 6.

Definition 1.5. For $\alpha \in [0, 1)$, let T_α be the linear mapping initially defined on sequences of compact support, given by

$$T_\alpha a(z) = e^{-i\pi\alpha} \sum_{n \in \mathbb{Z} \setminus \{0\}} (-1)^n a_n \Big(\frac{1}{z - \alpha - n} + \frac{1}{n} \Big) \frac{\sin[\pi(z - \alpha - n)]}{\pi} + e^{-i\pi\alpha} a_0 \operatorname{sinc}(z - \alpha). \tag{9}$$

For the spaces $\mathcal{X}_\alpha$ we prove the following results.

Theorem 3. *For $\alpha \in [0,1)$, T_α induces an isomorphism of Banach spaces $T_\alpha : \mathrm{BMO}(\mathbb{Z}) \to \mathcal{X}_\alpha$ (in particular, it is onto). Moreover, for each $L \in (\mathcal{B}^1_\pi)^*$ there exists a unique $f \in \mathcal{X}_\alpha$ that represents the functional L in the sense that for any $h \in \mathcal{B}^1_\pi$,*

$$L(f) = \sum_{n\in\mathbb{Z}} h(n+\alpha) f(n+\alpha). \tag{10}$$

It turns out that the spaces $\mathcal{X}_\alpha$, $\alpha \in [0,1)$, and $\mathrm{BMO}(e^{-i\pi z})$ they all coincide, thus providing a discrete characterization of the dual of $\mathcal{B}^1_\pi$.

Theorem 4. *For each $\alpha \in [0,1)$, $\mathrm{BMO}(e^{-i\pi z}) = \mathcal{X}_\alpha$, with equivalence of norms.*

Our next result shows that $\mathcal{B}^1_\pi$ is the dual space of a VMO-type space that can also be described in terms of compact Hankel operators. We recall that VMO is the subspace of BMO consisting of the closure of $C_0(\mathbb{R})$ in the BMO-norm – to be precise, the closure of $C_0(\mathbb{R})$ *modulo constants*. VMO can be charracterized as the subspace of BMO-functions φ such that

$$\lim_{\delta\to 0} \sup_{|I|<\delta} \frac{1}{|I|} \int_I |\varphi(x) - \varphi_I|\, dx = 0.$$

We also set $\mathrm{VMOA} := \mathrm{VMO} \cap \mathrm{BMOA}$.

Definition 1.6. We define $\mathrm{VMO}(e^{-i\pi z})$ to be the subspace of $\mathrm{BMO}(e^{-i\pi z})$ given by

$$\mathrm{VMO}(e^{-i\pi z}) = \left\{ f \in \mathcal{E}_\pi : \, e^{i\pi(\cdot)} f_0 \in \mathrm{VMOA}, \; e^{-i\pi(\cdot)} f_0 \in \overline{\mathrm{VMOA}} \right\} \Big/ \mathrm{span}\{e^{\pm i\pi(\cdot)}\},$$

with the norm of $\mathrm{BMO}(e^{-i\pi z})$.

Theorem 5. (1) *The projection P_π maps $C_0(\mathbb{R})$ onto $\mathrm{VMO}(e^{-i\pi z})$ and for all $f \in \mathrm{VMO}(e^{-i\pi z})$, $\|f\|_{\mathrm{BMO}(e^{-i\pi z})} \approx \inf\{\|\eta\|_{L^\infty} : \eta \in C_0(\mathbb{R}),\, P_\pi \eta = f\}$.*

(2) *The space $\mathcal{B}^1_\pi$ can be identified with $\mathrm{VMO}(e^{-i\pi z})^*$, the dual of $\mathrm{VMO}(e^{-i\pi z})$. Precisely, each $h \in \mathcal{B}^1_\pi$ defines a bounded linear functional on $\mathrm{VMO}(e^{-i\pi z})$ by the pairing*

$$L_h(\varphi) = \int_{\mathbb{R}} h(x)\varphi(x)\, dx = \int_{\mathbb{R}} h(x)\eta(x)\, dx$$

for any $\varphi \in \mathrm{VMO}(e^{-i\pi z})$, where $\varphi = P_\pi \eta$, $\eta \in C_0(\mathbb{R})$. Conversely, for each $L \in \mathrm{VMO}(e^{-i\pi z})^*$ there exists a unique $h \in \mathcal{B}^1_\pi$ such that $L = L_h$. Moreover, $\|L_h\|_{\mathrm{VMO}(e^{-i\pi z})^*} \approx \|h\|_{\mathcal{B}^1_\pi}$.

Theorem 6. *Let $\varphi \in \mathcal{E}_\pi$ be such that $\varphi_0 \in \mathcal{S}'$. Then, the following conditions are equivalent:*

1. *$\varphi \in \mathrm{VMO}(e^{-i\pi z})$;*
2. *the Hankel operator with symbol φ, $H_\varphi : \mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}$ is compact.*

The Paley–Wiener and Bernstein spaces constitute a cornerstone of function theory, analysis of holomorphic function spaces, signal analysis and operator theory. We point out that the spaces $\mathcal{B}^p_\kappa$ with $p \neq 2$ are often also called "Paley–Wiener spaces" in the literature. The classical Bernstein inequality [Ber23] (see also [PP37, And14, CP22a]) was first proved in the case of $\mathcal{B}^\infty_\kappa$. The extension to $\mathcal{B}^p_\kappa$ for $p < \infty$ (and $p \neq 2$) is simple. For this reason, the spaces $\mathcal{B}^p_\kappa$ are often called *Bernstein* spaces. In this work , we adhere to such terminology.

While in this paper we deal only with their 1-dimensional version, the higher dimensional generalizations have drawn a considerable amount of interest in recent times. We particularly refer to the papers [MPS21, MPS22, CP22a, CP22b], and references therein.

The paper is organized as follows. In Section 2 we prove some properties of the space $\mathrm{BMO}(e^{-i\pi z})$ and that the dual space $(\mathcal{B}^1_\pi)^*$ can be identified with $\mathrm{BMO}(e^{-i\pi z})$, and thus prove Theorem 1. In Section 3 we prove the characterization of $\mathrm{BMO}(e^{-i\pi z})$ in terms of boundedness of Hankel operators on $\mathcal{B}^2_{\pi/2}$, in particular we prove Theorem 2. Section 4 is devoted to the discrete characterization of the pairing of duality, along the lines of [Bes15]. In Section 5 we show that $\mathcal{B}^1_\pi$ is the dual space of a VMO-type space, whose elements are in turn precisely the symbols of compact Hankel operators on $\mathcal{B}^2_{\pi/2}$. We conclude with some remarks about extensions, open problems, and future works.

2. Proof of Theorem 1

We begin by collecting a few preliminary results. The first result can be proved exactly as [MPS21, Lemma 4.1]. That was stated for $p \in (1,\infty)$, but the proof applies *verbatim* also to the case $p = 1$.

Lemma 2.1. *The subspace $\mathcal{S}_\pi := \{f \in \mathcal{B}^1_\pi : f_0 \in \mathcal{S}\}$ is dense in $\mathcal{B}^1_\pi$.*

We are going to need the following, most likely known, simple lemma.

Lemma 2.2. *Let $F \in \mathcal{E}_\pi$ be such that $F(n) = 0$ for all $n \in \mathbb{Z}$, and that it satisfies condition (Y): $F(iy) = o\big(|y|e^{\pi|y|}\big)$. Then, there exists $a \in \mathbb{C}$ such that $F(z) = a\sin(\pi z)$.*

Proof. Let F be as in the statement. There exists an entire function G of order at most 1 such that $F(z) = \sin(\pi z)G(z)$. Denoting by I_f the indicator diagram of an entire function with respect to the order 1 we have, $I_F = [-i\pi, i\pi] + I_G$. By the properties of indicator diagrams (see [Lev96, Lectures 8-9]) we obtain that G is of exponential type 0, that is, $G \in \mathcal{E}_0$. Let us consider $G(z)/(z+1)$ on the right half-plane and $G(z)/(z-1)$ on the left half-plane. By applying the Phragmén–Lindelöf principle to these functions we see that G is a polynomial of degree at most 1, $G(z) = a + bz$. Therefore, $F(z) = a\sin(\pi z) + bz\sin(\pi z)$. But, if $b \neq 0$, F does not satisfy condition (Y), whence the conclusion. □

Theorem 2.3. *With the given norm, $\mathrm{BMO}(e^{-i\pi z})$ is a Banach space.*

Proof. We have observed that $\mathrm{BMO}(e^{-i\pi z})$ is well defined. In order to prove that the given one is a norm, we only need to prove the triangular inequality and that if $\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = 0$, then $f = 0$. Recall that

$$\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = \inf_{c_+\in\mathbb{C}} \|e^{i\pi(\cdot)}f_0 + c_+e^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} + \inf_{c_-\in\mathbb{C}} \|e^{-i\pi(\cdot)}f_0 + c_-e^{-i2\pi(\cdot)}\|_{\overline{\mathrm{BMOA}}}.$$

We claim that each of the two infima is actually a minimum. It suffices to consider the first one. Let m be such an infimum, and (c_k) be such that $\|e^{i\pi(\cdot)}f_0 + c_ke^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} \to m$ as $k \to \infty$. Then, (c_k) must be bounded, since otherwise $\|e^{i\pi(\cdot)}f_0 + c_ke^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} \geqslant \|c_ke^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} - \|e^{i\pi(\cdot)}f_0\|_{\mathrm{BMOA}}$, which tends to ∞ as $|c_k| \to \infty$. Then, we can assume $c_k \to c$ as $k \to \infty$. Then, the sequence of functions in $\mathrm{BMO}(\mathbb{R})$, $\big(e^{i\pi(\cdot)}f_0 + c_ke^{i2\pi(\cdot)}\big)_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathrm{BMO}(\mathbb{R})$, since

$$\big\|e^{i\pi(\cdot)}f_0 + c_ke^{i2\pi(\cdot)} - \big(e^{i\pi(\cdot)}f_0 + c_je^{i2\pi(\cdot)}\big)\big\|_{\mathrm{BMO}(\mathbb{R})} = |c_k - c_j|\|e^{i2\pi(\cdot)}\|_{\mathrm{BMO}(\mathbb{R})} \leqslant 2|c_k - c_j|.$$

This shows that each of the infima is a minimum. Therefore, there exist $c_{f,\pm} \in \mathbb{C}$ such that

$$\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = \|e^{i\pi(\cdot)}f_0 + c_{f,+}e^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} + \|e^{-i\pi(\cdot)}f_0 + c_{f,-}e^{-i2\pi(\cdot)}\|_{\overline{\mathrm{BMOA}}}. \tag{11}$$

Suppose that $\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = 0$. Then, $f_0 = -c_{f,\pm}e^{\pm i\pi(\cdot)}$ in $\mathrm{BMO}(\mathbb{R})$, so that

$$f = c_+ e^{i\pi(\cdot)} + c_- e^{-i\pi(\cdot)},$$

i.e. $f = 0$ in $\mathrm{BMO}(e^{-i\pi z})$. We observe in passing that the non-zero constans give rise to a non-trivial linear funcionals on $\mathcal{B}^1_\pi$, since $\widehat{f}(0) \neq 0$ in general.

Now, given $f, g \in \mathrm{BMO}(e^{-i\pi z})$, by (11) we have that

$$\begin{aligned}\|f+g\|_{\mathrm{BMO}(e^{-i\pi z})} &\leqslant \|e^{i\pi(\cdot)}(f_0+g_0) + (c_{f,+}+c_{g,+})e^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} \\ &\quad + \|e^{-i\pi(\cdot)}(f_0+g_0) + (c_{f,-}+c_{g,-})e^{-i2\pi(\cdot)}\|_{\overline{\mathrm{BMOA}}} \\ &\leqslant \|f\|_{\mathrm{BMO}(e^{-i\pi z})} + \|g\|_{\mathrm{BMO}(e^{-i\pi z})}.\end{aligned}$$

Finally, we check that $\mathrm{BMO}(e^{-i\pi z})$ is complete in its norm. Let $\{f_{(n)}\}$ be a Cauchy sequence in $\mathrm{BMO}(e^{-i\pi z})$. By the definition of the norm in $\mathrm{BMO}(e^{-i\pi z})$, see (7), it then follows that $\{e^{i\pi(\cdot)}f_{(n)}\}$ is a Cauchy sequence in $\mathrm{BMOA}\,/\,\mathrm{span}\{e^{i2\pi(\cdot)}\}$, and $\{e^{-i\pi(\cdot)}f_{(n)}\}$ is a Cauchy sequence in $\overline{\mathrm{BMOA}}/\,\mathrm{span}\{e^{-i2\pi(\cdot)}\}$. Such spaces are complete since, for instance, $\mathrm{BMOA}\,/\,\mathrm{span}\{e^{i2\pi(\cdot)}\}$ is is the dual space of $\{f \in H^1(\mathbb{C}^+) : \widehat{f}(2\pi) = 0\}$. Therefore, there exist $F \in \mathrm{BMOA}\,/\,\mathrm{span}\{e^{i2\pi(\cdot)}\}$ and $G \in \overline{\mathrm{BMOA}}/\,\mathrm{span}\{e^{-i2\pi(\cdot)}\}$ such that

$$e^{i\pi(\cdot)}f_{(n)} \to F \text{ in } \mathrm{BMOA}\,/\,\mathrm{span}\{e^{i2\pi(\cdot)}\}, \qquad e^{-i\pi(\cdot)}f_{(n)} \to G \text{ in } \overline{\mathrm{BMOA}}/\,\mathrm{span}\{e^{-i2\pi(\cdot)}\}$$

as $n \to \infty$. It follows that,

$$e^{i\pi(\cdot)}f_{(n)} \to F \text{ in } \mathcal{S}'/\,\mathrm{span}\{e^{i2\pi(\cdot)}, 1\}, \qquad e^{-i\pi(\cdot)}f_{(n)} \to G \text{ in } \mathcal{S}'/\,\mathrm{span}\{e^{-i2\pi(\cdot)}, 1\}.$$

Notice again that $\mathcal{S}'/\,\mathrm{span}\{e^{i2\pi(\cdot)}, 1\}$ is the dual of $\{\psi \in \mathcal{S} : \widehat{\psi}(2\pi) = \widehat{\psi}(0) = 0\}$, and similarly for $\mathcal{S}'/\,\mathrm{span}\{e^{-i2\pi(\cdot)}, 1\}$. The convergences imply that

$$f_{(n)} \to e^{-i\pi(\cdot)}F \text{ in } \mathcal{S}'/\,\mathrm{span}\{e^{i\pi(\cdot)}, e^{-i\pi(\cdot)}\}, \qquad f_{(n)} \to e^{i\pi(\cdot)}G \text{ in } \mathcal{S}'/\,\mathrm{span}\{e^{i\pi(\cdot)}, e^{-i\pi(\cdot)}\}.$$

We conclude that

$$e^{-i\pi(\cdot)}F = e^{i\pi(\cdot)}G =: \widetilde{f} \in \mathcal{S}'/\,\mathrm{span}\{e^{i\pi(\cdot)}, e^{-i\pi(\cdot)}\}. \tag{12}$$

Having constructed the limit f as an element of $\mathcal{S}'/\,\mathrm{span}\{e^{\pm i\pi(\cdot)}\}$, we want to show that $\widetilde{f}$ (in fact, any representative in its equivalence class) is the restriction of (an) $f \in \mathcal{E}_\pi$, that is, $\widetilde{f} = f_0$, where $f \in \mathrm{BMO}(e^{-i\pi z})$. Since $F \in \mathrm{BMOA}$ and $G \in \overline{\mathrm{BMOA}}$,

$$\mathrm{supp}(\widehat{e^{-i\pi(\cdot)}F}) \subseteq [-\pi, \infty), \qquad \mathrm{supp}(\widehat{e^{i\pi(\cdot)}G}) \subseteq (-\infty, \pi],$$

we have that

$$\mathrm{supp}(\widehat{\widetilde{f}}) \subseteq [-\pi, \pi].$$

The classical Paley–Wiener–Schwartz theorem (see e.g. [Tri10, Theorem 2, Chapter I]), $\widetilde{f}$ is indeed the restriction to $\mathbb{R}$ of a (unique) function $f \in \mathcal{E}_\pi$. By the definition of $\widetilde{f}$ (12) it is clear that $f \in \mathrm{BMO}(e^{-i\pi z})$. Finally,

$$\begin{aligned}\|f_{(n)} - f\|_{\mathrm{BMO}(e^{-i\pi z})} &= \inf_{c,d\in\mathbb{C}} \|e^{i\pi(\cdot)}(f_{(n)})_0 - F + ce^{i2\pi(\cdot)}\|_{\mathrm{BMOA}} + \|e^{-i\pi(\cdot)}(f_{(n)})_0 - G + de^{-i2\pi(\cdot)}\|_{\overline{\mathrm{BMOA}}} \\ &= \|e^{i\pi(\cdot)}(f_{(n)})_0 - F\|_{\mathrm{BMOA}\,/\,\mathrm{span}\{e^{i2\pi(\cdot)}\}} + \|e^{-i\pi(\cdot)}(f_{(n)})_0 - G\|_{\overline{\mathrm{BMOA}}/\,\mathrm{span}\{e^{-i2\pi(\cdot)}\}} \\ &\to 0,\end{aligned}$$

as $n \to \infty$. This completes the proof. □

Proof of Theorem 1. (1) We show that $P_\pi : L^\infty \to \mathrm{BMO}(e^{-i\pi z})$ is bounded. We adopt the following notation: Given an interval I, we denote by P_I the orthogonal projection of $L^2(\mathbb{R})$ onto the subspace of functions with support in I. If $I = [-\kappa, \kappa]$, we keep writing P_κ in place of $P_{[-\kappa,\kappa]}$. If $I = [0, \infty)$ ($I = (-\infty, 0]$, resp.), we write P_+ (P_-, resp.). We recall that $P_\pm : L^\infty \to \mathrm{BMOA}^\pm$, where, for later convenience, we write $\mathrm{BMOA}^+ = \mathrm{BMOA}$, and $\mathrm{BMOA}^- = \overline{\mathrm{BMOA}}$, are bounded and surjective, see e.g. [Gar07, Nik02].[4] We also recall that for $\varphi \in L^\infty(\mathbb{R})$, $P_+\varphi$ is defined as follows. Let $R > 0$ and $\overline{D}_R = \overline{D(0,R)} \subseteq \mathbb{C}$ be a fixed compact set, and let $I = \overline{D}_R \cap \mathbb{R}$, $I^* = \overline{D}_{3R} \cap \mathbb{R}$. Let $\varphi_1 = \varphi\chi_{I^*}$ and $\varphi_2 = \varphi - \varphi_1 = \varphi\chi_{(I^*)^c}$. Then, for $z \in \overline{D}_R$ given,

$$P_+\varphi(z) = \frac{1}{2\pi i}\,\mathrm{p.\,v.}\int_{I^*} \varphi(t)\frac{1}{t-z}\,dt + \frac{1}{2\pi i}\int_{(I^*)^c} \varphi(t)\Big[\frac{1}{t-z} - \frac{1}{t}\Big]\,dt. \tag{13}$$

We recall that $P_+\varphi$ is well defined on $\overline{D}_R$ up to an additive constant c.[5]

We first observe that on $L^2(\mathbb{R})$

$$\begin{aligned} P_\pi &= e^{-i\pi(\cdot)}P_{[0,2\pi]}(e^{i\pi(\cdot)}\cdot) = e^{-i\pi(\cdot)}\big(P_+ - P_{(2\pi,\infty)}\big)(e^{i\pi(\cdot)}\cdot) \\ &= e^{-i\pi(\cdot)}\Big(P_+ - e^{2i\pi(\cdot)}P_+\big(e^{-2i\pi(\cdot)}\cdot\big)\Big)(e^{i\pi(\cdot)}\cdot) \\ &= e^{-i\pi(\cdot)}P_+(e^{i\pi(\cdot)}\cdot) - e^{i\pi(\cdot)}P_+(e^{-i\pi(\cdot)}\cdot). \end{aligned} \tag{14}$$

Analogously,

$$P_\pi = e^{i\pi(\cdot)}P_-(e^{-i\pi(\cdot)}\cdot) - e^{-i\pi(\cdot)}P_-(e^{i\pi(\cdot)}\cdot). \tag{15}$$

Since $P_\pm$ extend to bounded linear operators from L^∞ into BMO, we may use (14), or equivalently (15), to define P_π on L^∞. If $\varphi \in L^\infty$, $R > 0$ and $z \in \overline{D}_R$, then using (13) we see that

$$\begin{aligned} P_\pi\varphi(z) &= e^{-i\pi z}P_+(e^{i\pi(\cdot)}\varphi)(z) - e^{i\pi z}P_+(e^{-i\pi(\cdot)}\varphi)(z) \\ &= \frac{1}{2\pi i}\,\mathrm{p.\,v.}\int_{I^*} \varphi(t)\frac{e^{i\pi(t-z)}}{t-z}\,dt + \frac{1}{2\pi i}\int_{(I^*)^c} \varphi(t)e^{i\pi(t-z)}\Big[\frac{1}{t-z} - \frac{1}{t}\Big]\,dt \\ &\quad - \frac{1}{2\pi i}\,\mathrm{p.\,v.}\int_{I^*} \varphi(t)\frac{e^{i\pi(z-t)}}{t-z}\,dt - \frac{1}{2\pi i}\int_{(I^*)^c} \varphi(t)e^{i\pi(z-t)}\Big[\frac{1}{t-z} - \frac{1}{t}\Big]\,dt \\ &= \int_{I^*} \varphi(t)\,\mathrm{sinc}(t-z)\,dt + \int_{(I^*)^c} \varphi(t)\frac{\sin(\pi(t-z))}{\pi}\Big[\frac{1}{t-z} - \frac{1}{t}\Big]\,dt. \end{aligned} \tag{16}$$

It is simple to check that $P_+\varphi$ is defined on each closed strip $\{z : |\operatorname{Re} z| \leqslant R\}$, modulo a constant.

Notice that, since for $\varphi \in L^\infty$, $P_\pm\varphi$ is defined only up to an additive constant, $P_\pi\varphi$ is defined only modulo $\operatorname{span}\{e^{i\pi(\cdot)}, e^{-i\pi(\cdot)}\}$. It is then clear that P_π maps L^∞ into $\mathcal{E}_\pi/\operatorname{span}\{e^{i\pi(\cdot)}, e^{-i\pi(\cdot)}\}$.

We now observe that the boundary values of functions in $H^\infty(\mathbb{C}^+)$ are multipliers of BMOA, since they are bounded multipliers of $H^1(\mathbb{C}^+)$. In particular, $e^{\pm i2\pi(\cdot)}$ is a multiplier of $\mathrm{BMOA}^\pm$, resp., with norm $\leqslant 1$.

[4] Notice that, when acting on $\mathcal{S}'$, $P_{[0,\infty)} = P_{(0,\infty)}$ modulo polynomials. Hence, when acting on L^∞ the two operators differ by a constant, hence coincide having values in BMOA.

[5] For, if $R' > R$, and $P_+\varphi$ is defined for $z \in \overline{D}_{R'}$ as in (13), then on $\overline{D}_R$ the two definitions differ by the constant $c = \int_{I^*_{R'}\setminus I^*_R} \varphi(t)/t\,dt$, where $I^*_{R'} = [-3R', 3R']$.

Let $\varphi \in L^\infty$. Then, using (14) we see that

$$\begin{aligned}\|e^{i\pi(\cdot)}P_\pi\varphi\|_{\mathrm{BMO}} &\leqslant \|P_+(e^{i\pi(\cdot)}\varphi)\|_{\mathrm{BMO}} + \|e^{2i\pi(\cdot)}P_+(e^{-i\pi(\cdot)}\varphi)\|_{\mathrm{BMO}}\\ &\leqslant \|e^{i\pi(\cdot)}\varphi\|_{L^\infty} + \|e^{-i\pi(\cdot)}\varphi\|_{L^\infty} = 2\|\varphi\|_{L^\infty}.\end{aligned}$$

Using (15) we obtain that $\|e^{-i\pi(\cdot)}P_\pi\varphi\|_{\mathrm{BMO}} \leqslant 2\|\varphi\|_{L^\infty}$ so that,

$$\begin{aligned}\|P_\pi\varphi\|_{\mathrm{BMO}(e^{-i\pi z})} &\leqslant \|e^{i\pi(\cdot)}P_\pi\varphi\|_{\mathrm{BMO}} + \|e^{-i\pi(\cdot)}P_\pi\varphi\|_{\mathrm{BMO}}\\ &\leqslant 4\|\varphi\|_{L^\infty}.\end{aligned}$$

Hence, $P_\pi : L^\infty \to \mathrm{BMO}(e^{-i\pi z})$ is bounded. In order to show that it is surjective, let $f \in \mathrm{BMO}(e^{-i\pi z})$. Then, $e^{\pm i\pi(\cdot)}f_0 \in \mathrm{BMOA}^\pm$, so that there exist $\varphi_\pm \in L^\infty$ such that

$$e^{i\pi(\cdot)}f_0 = P_+\varphi_+, \quad \text{and} \quad e^{-i\pi(\cdot)}f_0 = P_-\varphi_-,. \tag{17}$$

Therefore,

$$f_0 = e^{-i\pi(\cdot)}P_+\varphi_+ = P_{[-\pi,\infty)}(e^{-i\pi(\cdot)}\varphi_+) \quad \text{and also} \quad f_0 = e^{i\pi(\cdot)}P_-\varphi_- = P_{(-\infty,\pi]}(e^{i\pi(\cdot)}\varphi_-).$$

However, $\operatorname{supp}(\widehat{f_0}) \subseteq [-\pi,\pi]$, so that the above equalities imply that $\operatorname{supp}(\widehat{e^{-i\pi(\cdot)}\varphi_+}) \subseteq (-\infty,\pi]$, and $\operatorname{supp}(\widehat{e^{i\pi(\cdot)}\varphi_-}) \subseteq [-\pi,\infty)$. Therefore,

$$f_0 = P_\pi(e^{-i\pi(\cdot)}\varphi_+) = P_\pi(e^{i\pi(\cdot)}\varphi_-);$$

hence $P_\pi : L^\infty \to \mathrm{BMO}(e^{-i\pi z})$ is onto.

Next, since $P_\pi : L^\infty \to \mathrm{BMO}(e^{-i\pi z})$ is bounded,

$$\|f\|_{\mathrm{BMO}(e^{-i\pi z})} \leqslant 4\inf\{\|\varphi\|_{L^\infty} : \varphi \in L^\infty,\, P_\pi\varphi = f\}.$$

On the other hand, if $g \in \mathrm{BMOA}$, then $\|g\|_{\mathrm{BMOA}} \approx \inf\{\|\varphi\|_{L^\infty} : \varphi \in L^\infty,\, P_+\varphi = g\}$, and analogously if $g \in \overline{\mathrm{BMOA}}$. Therefore, since $e^{i\pi(\cdot)}f_0 + c_{f,+}e^{i2\pi(\cdot)} \in \mathrm{BMOA}$, and $e^{-i\pi(\cdot)}f_0 + c_{f,-}e^{-i2\pi(\cdot)} \in \overline{\mathrm{BMOA}}$, there exist a constant $C_1 > 0$ and $\varphi_\pm \in L^\infty$ such that (17) holds and such that $\|\varphi_\pm\|_{L^\infty} \leqslant C_1\|e^{\pm i\pi(\cdot)}f_0 + c_{f,\pm}e^{\pm i2\pi(\cdot)}\|_{\mathrm{BMO}(\mathbb{R})}$. Then,

$$\|e^{i\pi(\cdot)}\varphi_+\|_{L^\infty} = \|\varphi_+\|_{L^\infty} \leqslant C_1\|e^{i\pi(\cdot)}f_0 + c_{f,+}e^{i2\pi(\cdot)}\|_{\mathrm{BMO}(\mathbb{R})} \leqslant C_1\|f\|_{\mathrm{BMO}(e^{-i\pi z})},$$

as we wished to show.

We now turn to the identification of $(\mathcal{B}^1_\pi)^*$ with $\mathrm{BMO}(e^{-i\pi z})$. Let $f \in \mathrm{BMO}(e^{-i\pi z})$, and let $\varphi \in L^\infty$ such that $f = P_\pi\varphi$ and $\|\varphi\|_{L^\infty} \leqslant C_1\|f\|_{\mathrm{BMO}(e^{-i\pi z})}$. Then, letting L_f be initially defined on $\mathcal{S}_\pi$, we have that

$$\begin{aligned}|L_f(h)| &= |\int_{\mathbb{R}} h(x)f(x)\,dx| = |\int_{\mathbb{R}} h(x)P_\pi\varphi(x)\,dx| = |\langle h, P_\pi\varphi\rangle|\\ &= |\langle\widehat{h(-\cdot)}, \widehat{P_\pi\varphi}\rangle| = |\langle\widehat{h(-\cdot)}, \widehat{\varphi}\rangle|\\ &\leqslant \|h\|_{L^1}\|\varphi\|_{L^\infty} \leqslant C_1\|h\|_{\mathcal{B}^1_\pi}\|f\|_{\mathrm{BMO}(e^{-i\pi z})},\end{aligned}$$

so that $\|L_f\|_{(\mathcal{B}^1_\pi)^*} \leqslant C_1\|f\|_{\mathrm{BMO}(e^{-i\pi z})}$.

Conversely, if $L \in (\mathcal{B}^1_\pi)^*$, by Hahn–Banach there exists $\varphi \in L^\infty$ such that $L(h) = \int_{\mathbb{R}} h\varphi\,dm$ for all $h \in \mathcal{B}^1_\pi$, with $\|\varphi\|_{L^\infty} = \|L\|_{(\mathcal{B}^1_\pi)^*}$. However,

$$\int_{\mathbb{R}} h(x)\varphi(x)\,dx = \int_{\mathbb{R}} h(x)P_\pi(\varphi(-\cdot))(-x)\,dx = \int_{\mathbb{R}} h(x)P_\pi\varphi(x)\,dx =: \int_{\mathbb{R}} h(x)f(x)\,dx$$

where $f = P_\pi \varphi \in \mathrm{BMO}(e^{-i\pi z})$. Then, $L = L_f$. Moreover,

$$\|f\|_{\mathrm{BMO}(e^{-i\pi z})} = \|P_\pi \varphi\|_{\mathrm{BMO}(e^{-i\pi z})} \leqslant 4\|\varphi\|_{L^\infty} = 4\|L_f\|_{(\mathcal{B}^1_\pi)^*}.$$

This proves the theorem. □

3. Proof of Theorem 2

Recall the definition (6) of Hankel operator H_φ on $\mathcal{B}^2_\kappa$, having as symbol the locally integrable function φ which is also a tempered distribution on $\mathbb{R}$. Notice that H_φ is well defined on the dense subset $\mathcal{S}_\kappa$ of $\mathcal{B}^2_\kappa$. We wish to characterize the symbols φ for which H_φ extends to a bounded linear operator on $\mathcal{B}^2_\kappa$. For general facts about Hankel operators on Hardy spaces we refer to [Nik02].

The space $\mathcal{B}^2_\kappa$ is endowed with the natural involution $f \mapsto f^\#$, where $f^\#(z) = \overline{f(\overline{z})}$. Hence, for $x \in \mathbb{R}$, $f_0^\#(x) = \overline{f_0(x)}$. Then, for $f \in \mathcal{B}^2_\kappa$, arguing formally for the time being, we have that

$$\begin{aligned} H_\varphi f(z) &= \mathcal{F}^{-1}\big(\chi_{[-\kappa,\kappa]}\mathcal{F}(\varphi f_0^\#)\big)(z) \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\kappa}^{\kappa} e^{iz\xi}(\widehat{\varphi} * \widehat{f_0^\#})(\xi)\, d\xi \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\kappa}^{\kappa} e^{iz\xi}\int_{-\kappa}^{\kappa} \widehat{\varphi}(\xi - t)\widehat{f_0^\#}(t)\, dt\, d\xi. \end{aligned}$$

Therefore, the operator H_φ remains unchanged if we replace φ with $\mathcal{F}^{-1}\big(\chi_{[-2\kappa,2\kappa]}\widehat{\varphi}\big)$. Therefore, we may assume, and we do so in what follows, that $\mathrm{supp}(\widehat{\varphi}) \subseteq [-2\kappa, 2\kappa]$.[6]

Let now $\eta_L \in C^\infty_c(\mathbb{R})$ be such that $\mathrm{supp}\,\eta \subseteq [-4\kappa, -\kappa/2]$, $\eta \geqslant 0$, and $\eta = 1$ on $[-3\kappa, -\kappa]$. Let $\eta_R = \eta_L(-\cdot)$ and $\eta_C = \chi_{[-2\kappa,2\kappa]} - \eta_L - \eta_R$. Define then $\varphi_L = \mathcal{F}^{-1}(\eta_L\widehat{\varphi})$, and analogously $\varphi_R = \mathcal{F}^{-1}(\eta_R\widehat{\varphi})$ and $\varphi_C = \mathcal{F}^{-1}(\eta_C\widehat{\varphi})$. We say that a symbol ψ is *analytic* if $\widehat{\psi}$ is supported in $[0,\infty)$, and analogously, that it is *anti-analytic* if $\widehat{\psi}$ is supported in $(-\infty, 0]$. Then, φ_L is anti-analytic and φ_R analytic.

In [Roc87, Theorem 4.1] it is proved that H_φ is bounded if and only if

- $P_-\big(e^{-2i\kappa(\cdot)}\varphi_L\big) \in \mathrm{BMO}(\mathbb{R})$;
- $P_+\big(e^{2i\kappa(\cdot)}\varphi_R\big) \in \mathrm{BMO}(\mathbb{R})$;
- $\varphi_C \in L^\infty(\mathbb{R})$.

Moreover,

$$\|H_\varphi\|_{\mathcal{B}^2_\kappa \to \mathcal{B}^2_\kappa} \approx \|P_-\big(e^{-2i\kappa(\cdot)}\varphi_L\big)\|_{\mathrm{BMO}} + \|P_+\big(e^{2i\kappa(\cdot)}\varphi_R\big)\|_{\mathrm{BMO}} + \|\varphi_C\|_{L^\infty}.$$

Another key fact we are going to use is a factorization decomposition of $\mathcal{B}^1_\kappa$, obtained by A. Baranov, R. Bessonov, and V. Kapustin [BBK11, Proposition 4.1, and Section 7] (as corollary of a more general result). Given any $h \in \mathcal{B}^1_{2\kappa}$, there exist $f_j, g_j \in \mathcal{B}^2_\kappa$, $j = 1, \dots, 4$, such that $h = \sum_{j=1}^4 f_j g_j$ and

$$\|h\|_{\mathcal{B}^1_{2\kappa}} \approx \inf\Big\{ \sum_{j=1}^4 \|f_j\|_{\mathcal{B}^2_\kappa}\|g_j\|_{\mathcal{B}^2_\kappa} : h = \sum_{j=1}^4 f_j g_j \Big\}. \tag{18}$$

It is easy to see that if $h \in \mathcal{S}_{2\kappa}$, it is possible to select $f_j, g_j \in \mathcal{S}_\kappa$ such that (18) holds.

[6] We recall that it is not in general true that $H_\varphi = H_{P_\kappa\varphi}$ since $\mathcal{E}_\kappa$ is not an algebra.

Proof of Theorem 2. We are going to use the above mentioned results with $\kappa = \pi/2$. Observe that if φ is as in (2), since we are assuming, as we may, that $\operatorname{supp}(\widehat{\varphi}) \subseteq [-\pi, \pi]$, there exists $f \in \mathcal{E}_\pi$ such that $\varphi = f_0$. Notice that Theorem 1 shows that (1) and (2) are equivalent.

(3) $\Rightarrow$ (2). Given $h \in \mathcal{S}_\pi$, let $h = \sum_{j=1}^4 f_j g_j$ be as in (18), with $f_j, g_j \in \mathcal{B}^2_{\pi/2}$. By Lemma 2.1, without loss of generality we may assume f_j, g_j to be in $\mathcal{S}_{\pi/2}$. Then,

$$\begin{aligned}\langle h, \varphi\rangle &= \Big\langle \sum_{j=1}^4 f_j g_j, \varphi \Big\rangle \\ &= \sum_{j=1}^4 \int_{\mathbb{R}} f_j(x) g_j(x) \varphi(x)\, dx = \sum_{j=1}^4 \int_{\mathbb{R}} f_j(x) \overline{P_{\pi/2}(\overline{g_j \varphi})(x)}\, dx \\ &= \sum_{j=1}^4 \int_{\mathbb{R}} f_j(x) \overline{H_{\overline{\varphi}} g_j(x)}\, dx.\end{aligned}$$

Therefore,

$$|\langle h, \varphi\rangle| \leqslant \sum_{j=1}^4 \|f_j\|_{\mathcal{B}^2_{\pi/2}} \|H_{\overline{\varphi}}\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}} \|g_j\|_{\mathcal{B}^2_{\pi/2}} \leqslant C_2 \|h\|_{\mathcal{B}^1_\pi} \|H_{\overline{\varphi}}\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}},$$

so that $\varphi \in (\mathcal{B}^1_\pi)^*$, and moreover $\|\varphi\|_{(\mathcal{B}^1_\pi)^*} \leqslant C_2 \|H_{\overline{\varphi}}\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}}$. We observe that

$$H_{\overline{\varphi}} f = H_\varphi f^\#$$

and since $f \mapsto f^\#$ is an isometry on the Paley–Wiener (and Bernstein) spaces, $H_{\overline{\varphi}}$ is bounded if and only if H_φ is, with equality of norms.

(1) $\Rightarrow$ (3). If $f, g \in \mathcal{B}^2_{\pi/2}$, then

$$\begin{aligned}|\langle H_\varphi f \,|\, g\rangle| = |\langle \varphi \,|\, fg\rangle| = |\langle P_\pi \widetilde{\varphi} \,|\, fg\rangle| = |\langle \widetilde{\varphi} \,|\, fg\rangle| &\leqslant \|\widetilde{\varphi}\|_{L^\infty} \|fg\|_{\mathcal{B}^1_\pi} \\ &\leqslant \|\widetilde{\varphi}\|_{L^\infty} \|f\|_{\mathcal{B}^2_{\pi/2}} \|g\|_{\mathcal{B}^2_{\pi/2}}.\end{aligned}$$

Therefore, H_φ is bounded on $\mathcal{B}^2_{\pi/2}$ and $\|H_\varphi\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}} \leqslant \inf\{\|\widetilde{\varphi}\|_{L^\infty} : \varphi = P_\pi \widetilde{\varphi},\, \widetilde{\varphi} \in L^\infty\}$.

Finally, we have shown that

$$\|\varphi\|_{(\mathcal{B}^1_\pi)^*} \leqslant C_2 \|H_\varphi\|_{\mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}} \leqslant C_2 \inf\{\|\widetilde{\varphi}\|_{L^\infty} : \varphi = P_\pi \widetilde{\varphi},\, \widetilde{\varphi} \in L^\infty\}.$$

From Theorem 1 we also have that $\varphi = f_0$ for some $f \in \mathrm{BMO}(e^{-i\pi z})$ and

$$\|\varphi\|_{(\mathcal{B}^1_\pi)^*} \approx \|f\|_{\mathrm{BMO}(e^{-i\pi z})} \approx \inf\{\|\widetilde{\varphi}\|_{L^\infty} : f = P_\pi \widetilde{\varphi},\, \widetilde{\varphi} \in L^\infty\}.$$

The stated norm equivalences now follow. □

4. Discrete nature of the pairing of duality

We now turn to the discrete characterization of $(\mathcal{B}^1_\pi)^*$. We begin with some properties of $\mathcal{X}_\alpha$.

Proposition 4.1. *For $\alpha \in [0, 1)$, $\mathcal{X}_\alpha$ is a Banach space.*

Proof. We have observed that $\mathcal{X}_\alpha$ is well defined. In order to prove that the given one is a norm, we only need to prove the triangular inequality and that if $\|f\|_{\mathcal{X}_\alpha} = 0$, then $f = 0$.

Observe that $\operatorname{span}\{e^{\pm i\pi(\cdot)}\} = \operatorname{span}\{\cos\pi(\cdot), \sin\pi(\cdot)\}$ and that

$$\begin{aligned}\|f\|_{\mathcal{X}_\alpha} &= \inf_{c_\pm\in\mathbb{C}} \big\|e^{i\pi(\cdot+\alpha)}\big(f(\cdot+\alpha) + c_+e^{i\pi(\cdot)} + c_-e^{-i\pi(\cdot)}\big)\big\|_{\mathrm{BMO}(\mathbb{Z})}\\ &= \big\|(-1)^{(\cdot)}\big(f(\cdot+\alpha)\big)\big\|_{\mathrm{BMO}(\mathbb{Z})}. \end{aligned} \tag{19}$$

Suppose that $\|f\|_{\mathcal{X}_\alpha} = 0$. Then, $\big((-1)^{(\cdot)}\big(f(\cdot+\alpha)\big)\big) = 0$ in $\mathrm{BMO}(\mathbb{Z})$. It is then easy to see that (for every representative of f in its equivalence class), there exist constants $a, b \in \mathbb{C}$ such that if we set

$$F(z) := f(z+\alpha) + a\cos(\pi z) + b\sin(\pi z)$$

then F vanishes at all points $n \in \mathbb{Z}$. Moreover, any such F satisfies condition (Y) of Lemma 2.2 and therefore $F(z) = c\sin(\pi z)$ for some $c \in \mathbb{C}$. This implies that $f(\cdot+\alpha) \in \operatorname{span}\{e^{\pm i(\cdot)}\}$, that is, $f = 0$ in $\mathcal{X}_\alpha$.

Finally, given $f, g \in \mathcal{X}_\alpha$, by (19) we have that

$$\begin{aligned}\|f+g\|_{\mathcal{X}_\alpha} &= \big\|(-1)^{(\cdot)}\big(f(\cdot+\alpha) + g(\cdot+\alpha)\big)\big\|_{\mathrm{BMO}(\mathbb{Z})}\\ &\leqslant \big\|(-1)^{(\cdot)}\big(f(\cdot+\alpha)\big)\big\|_{\mathrm{BMO}(\mathbb{Z})} + \big\|(-1)^{(\cdot)}g(\cdot+\alpha)\big)\big\|_{\mathrm{BMO}(\mathbb{Z})}\\ &= \|f\|_{\mathcal{X}_\alpha} + \|g\|_{\mathcal{X}_\alpha}.\end{aligned}$$

This proves the Proposition. □

Recall that, see e.g. [CW77], if $a \in \mathrm{BMO}(\mathbb{Z})$, then

$$\sum_{n\in\mathbb{Z}} \frac{|a_n|}{1+n^2} < \infty. \tag{20}$$

Proof of Theorem 3. Step 1. The mapping T_α is well defined on $\mathrm{BMO}(\mathbb{Z})$ and if $a \in \mathrm{BMO}(\mathbb{Z})$, $T_\alpha a$ is an entire function.

Let $\mathbf{1} = (c_n)_{n\in\mathbb{Z}}$ be the constant sequence with $c_n = 1$ for all $n \in \mathbb{Z}$. Then, we have

$$\begin{aligned}e^{i\pi\alpha}T_\alpha\mathbf{1}(z) &= \sum_{n\in\mathbb{Z}\setminus\{0\}} (-1)^n\Big(\frac{1}{z-\alpha-n} + \frac{1}{n}\Big)\frac{\sin[\pi(z-\alpha-n)]}{\pi} + \operatorname{sinc}(z-\alpha)\\ &= e^{i\pi\alpha}\cos(\pi(z-\alpha)),\end{aligned}$$

by a standard identity, see e.g. [Lan99, Example 2.4, Ch. XIII]. Hence, $T_\alpha\mathbf{1} = 0$ in $\mathcal{X}_\alpha$, that is, T_α is well defined on $\mathrm{BMO}(\mathbb{Z})$.

Let now $a = (a_n) \in \mathrm{BMO}(\mathbb{Z})$, and choose the representative sequence such that $a_0 = 0$. Fix $R > 1$ and the compact set $\overline{D(0,R)}$. Then,

$$\begin{aligned}e^{i\pi\alpha}T_\alpha a(z) &= \sum_{n\in\mathbb{Z}\setminus\{0\}} a_n\Big(\frac{1}{z-\alpha-n} + \frac{1}{n}\Big)\frac{\sin[\pi(z-\alpha)]}{\pi}\\ &= \Big(\sum_{0<|n|\leqslant 2R} + \sum_{|n|>2R}\Big)a_n\frac{z-\alpha}{(z-\alpha-n)n}\frac{\sin[\pi(z-\alpha)]}{\pi}\\ &=: S_1(z) + S_2(z).\end{aligned}$$

For $z \in \overline{D(0,R)}$ and $|n| > 2R$ we observe that

$$\begin{aligned}\Big|a_n \frac{z-\alpha}{(z-\alpha-n)n} \frac{\sin[\pi(z-\alpha)]}{\pi}\Big| &\leqslant \sup_{|z|\leqslant 2R} \frac{1}{\pi}\big|(z-\alpha)\sin[\pi(z-\alpha)]\big| \frac{|a_n|}{|n|(|n|-R-1)|} \\ &\leqslant C_R \frac{|a_n|}{|n|(|n|+1)},\end{aligned}$$

so that the sum defining S_2 converges uniformly on $\overline{D(0,R)}$. On the other hand, if $z \in \overline{D(0,R)}$ and $|n| \leqslant 2R$ we have that

$$\begin{aligned}\Big|a_n \frac{z-\alpha}{(z-\alpha-n)n} \frac{\sin[\pi(z-\alpha)]}{\pi}\Big| &= |z-\alpha||\operatorname{sinc}(z-\alpha-n)| \frac{|a_n|}{|n|} \\ &\leqslant C'_R \frac{|a_n|}{|n|}\end{aligned}$$

and since S_1 is given by a finite sum, the convergence is uniform. Thus, $T_\alpha a$ defines an entire function.

Step 2. For $a \in \mathrm{BMO}(\mathbb{Z})$, $T_\alpha a \in \mathcal{E}_\pi$.

We slipt the sum defining $T_\alpha a$ as follows:

$$\begin{aligned}e^{i\pi\alpha} T_\alpha a(z) &= \Big(\sum_{0<|n|\leqslant 2|z|} + \sum_{|n|>2|z|} \Big) a_n \frac{z-\alpha}{(z-\alpha-n)n} \frac{\sin[\pi(z-\alpha)]}{\pi} \\ &=: \widetilde{S}_1(z) + \widetilde{S}_2(z).\end{aligned}$$

Using (20) we see that

$$\begin{aligned}|\widetilde{S}_2(z)| &\leqslant \sum_{|n|>2|z|} \frac{1}{\pi}\big|(z-\alpha)\sin[\pi(z-\alpha)]\big| \frac{|a_n|}{|n|(|n|-R-1)|} \\ &\leqslant C_R \frac{1}{\pi}\big|(z-\alpha)\sin[\pi(z-\alpha)]\big| \sum_{|n|>2|z|} \frac{|a_n|}{|n|(|n|+1)|} \\ &\leqslant C'_R \frac{1}{\pi}\big|(z-\alpha)\sin[\pi(z-\alpha)]\big|.\end{aligned}$$

Hence, $\widetilde{S}_2 \in \mathcal{E}_\pi$. Next, writing $z = x + iy$,

$$\begin{aligned}|\widetilde{S}_1(z)| &\leqslant \sum_{0<|n|\leqslant 2|z|} |z-\alpha||\operatorname{sinc}(z-\alpha-n)| \frac{|a_n|}{|n|} \\ &\leqslant e^{\pi|y|}|z-\alpha| \sum_{0<|n|\leqslant 2|z|} 2|z| \frac{|a_n|}{|n|^2}.\end{aligned}$$

This shows that $T_\alpha a \in \mathcal{E}_\pi$.

Step 3. We show that for $a \in \mathrm{BMO}(\mathbb{Z})$, $T_\alpha a(iy) = o(|y|e^{\pi|y|})$.

We have

$$\lim_{|y|\to\infty} \frac{|T_\alpha a(iy)|}{|y|e^{\pi|y|}} = \lim_{|y|\to\infty} \frac{|iy-\alpha||\operatorname{sinc}(z-\alpha-n)|}{|y|e^{\pi|y|}} \Big| \sum_{n\neq 0} \frac{a_n}{(iy-\alpha-n)n}\Big|,$$

so it suffices to show that

$$\lim_{|y|\to\infty} \sum_{n\neq 0} \frac{a_n}{(iy-\alpha-n)n} = 0.$$

This follows at once from the dominated convergence theorem, whence $T_\alpha a(iy) = o(|y|e^{\pi|y|})$.

Step 4. We now show that $T_\alpha a \in \mathcal{X}_\alpha$.

When $k \neq 0$ we have that

$$\begin{aligned} & e^{i\pi(k+\alpha)} T_\alpha a(k+\alpha) \\ & = (-1)^k \Bigg(\sum_{n\in\mathbb{Z}\setminus\{0,k\}} (-1)^n a_n \Big(\frac{1}{k-n} + \frac{1}{n}\Big) \frac{\sin[\pi(k-n)]}{\pi} + (-1)^k a_k + (-1)^k a_0 \operatorname{sinc}(k) \Bigg) \\ & = a_k. \end{aligned} \tag{21}$$

The case $k = 0$ is immediate, and therefore

$$\big(e^{i\pi(\cdot+\alpha)} T_\alpha a(\cdot+\alpha)\big)_{|\mathbb{Z}} = a \in \mathrm{BMO}(\mathbb{Z}).$$

This fact, together with the conclusions of *Steps 1-3* imply that $T_\alpha \in \mathcal{X}_\alpha$, as we wished to show.

Step 5. The operator T_α is an isomorphism.

Suppose $T_\alpha a = 0$, that is, $T_\alpha a(z) = c_1 \cos(\pi(z-\alpha)) + c_2 \sin(\pi(z-\alpha))$, for some $c_1, c_2 \in \mathbb{C}$. In *Step 1* we have observed that $T(c_1\mathbf{1}) = c_1 \cos(\pi(z-\alpha))$. Since, in $\mathrm{BMO}(\mathbb{Z})$, $a = a + c_1\mathbf{1}$, we have that $T_\alpha a(z) = c_2 \sin(\pi(z-\alpha))$. By (21) it then follows that

$$a_k = e^{i\pi(k+\alpha)} T_\alpha a(k+\alpha) = 0.$$

Hence, $a_k = 0$ for all $k \in \mathbb{Z}$ so that the equality holds with $c_2 = 0$ and T_α is injective.

Next, let $f \in \mathcal{X}_\alpha$. Then $a := \big((-1)^{(\cdot)} f(\cdot+\alpha)\big) \in \mathrm{BMO}(\mathbb{Z})$. Therefore, $f - T_\alpha a \in \mathcal{X}_\alpha$ and $f(n+\alpha) - T_\alpha a(n) = 0$ for every $n \in \mathbb{Z}$. By Lemma 2.2, there exists $c \in \mathbb{C}$ such that $f = T_\alpha a + c \sin \pi(\cdot)$, that is, $T_\alpha a = f$ in $\mathcal{X}_\alpha$, i.e. T_α is also onto.

Step 6. We now wish to show that each of the spaces $\mathcal{X}_\alpha$, $\alpha \in [0,1)$ can be identified with the dual space $(\mathcal{B}^1_\pi)^*$.

From [Eof95] we know that $\widetilde{T}_\alpha : H^1(\mathbb{Z}) \to \mathcal{B}^1_\pi$ is an isomorphism, where

$$\widetilde{T}_\alpha a(z) = \sum_{n\in\mathbb{Z}} a_n (-1)^n \operatorname{sinc}(z-\alpha-n).$$

where $\widetilde{T}_\alpha^{-1}$ is given by

$$\widetilde{T}_\alpha^{-1} f = \big(e^{i\pi\alpha}(-1)^{(\cdot)} f(\cdot+\alpha)\big)_{|\mathbb{Z}}.$$

Then, $(\widetilde{T}_\alpha^{-1})^* : \mathrm{BMO}(\mathbb{Z}) \to (\mathcal{B}^1_\pi)^*$ is an isomorphism, where, if $f \in \mathcal{B}^1_\pi$ and $b \in \mathrm{BMO}(\mathbb{Z})$, the pairing of duality is given by

$$\begin{aligned} (\widetilde{T}_\alpha^{-1})^* b(f) &= \langle \widetilde{T}_\alpha^{-1} f, b\rangle = \langle b, \big(e^{i\pi\alpha}(-1)^{(\cdot)} f(\cdot+\alpha)\big)_{|\mathbb{Z}}\rangle \\ &= \langle \big(e^{i\pi\alpha}(-1)^{(\cdot)} T_\alpha b(\cdot+\alpha)\big)_{|\mathbb{Z}}, \big(e^{i\pi\alpha}(-1)^{(\cdot)} f(\cdot+\alpha)\big)_{|\mathbb{Z}}\rangle \\ &= \langle e^{2i\alpha\pi} \big(T_\alpha b(\cdot+\alpha)\big)_{|\mathbb{Z}}, \big(f(\cdot+\alpha)\big)_{|\mathbb{Z}}\rangle \end{aligned}$$

This proves the theorem. □

Corollary 4.2. *For each $\alpha \in (0,1)$, the space $\mathcal{X}_\alpha$ coincides with $\mathcal{X}_0$, with equivalence of norms. Hence, the spaces $\mathcal{X}_\alpha$ are all equal.*

Let $f \in \mathcal{X}_0$. In *Step 6* we have shown that $T_0 a = f$, where $a := \big((-1)^{(\cdot)} f(\cdot)\big) \in \mathrm{BMO}(\mathbb{Z})$, so that

$$\begin{aligned}(-1)^k f(k+\alpha) &= (-1)^k \Big(\sum_{n\neq 0} (-1)^n a_n \Big(\frac{1}{k+\alpha-n} + \frac{1}{n} \Big) \frac{\sin(\pi(k+\alpha-n))}{\pi} + a_0 \frac{\sin(\pi(k+\alpha))}{\pi(k+\alpha)} \Big) \\ &= (-1)^k \frac{\sin(\pi(k+\alpha))}{\pi} \sum_{n\neq 0} a_n \Big(\frac{1}{k+\alpha-n} + \frac{1}{n} \Big) + (-1)^k a_0 \operatorname{sinc}(k+\alpha) \\ &= \frac{e^{i\pi\alpha}}{\pi} \big(H_{d,\alpha} a\big)_k + (-1)^k a_0 \operatorname{sinc}(k+\alpha).\end{aligned}$$

Therefore, possibly replacing f by $f - a_0 \operatorname{sinc}(\cdot)$, or assuming that $a_0 = 0$ as we also may, we obtain that

$$R_\alpha f := e^{-i\pi\alpha} \big((-1)^{(\cdot)} f(\cdot + \alpha)\big) = \frac{1}{\pi} H_{d,\alpha} R_0 f \in \mathrm{BMO}(\mathbb{Z}).$$

Since $H_{d,\alpha} : \mathrm{BMO}(\mathbb{Z}) \to \mathrm{BMO}(\mathbb{Z})$ is an isomorphism, it follows that $\big((-1)^{(\cdot)} f(\cdot+\alpha)\big) \in \mathrm{BMO}(\mathbb{Z})$, and

$$\begin{aligned}\|f\|_{\mathcal{X}_\alpha} &\leqslant \big\| e^{-i\pi\alpha} \big((-1)^{(\cdot)} f(\cdot+\alpha)\big) \big\|_{\mathrm{BMO}(\mathbb{Z})} \\ &\leqslant C_\alpha \|R_0 f\|_{\mathrm{BMO}(\mathbb{Z})} \\ &\leqslant C_\alpha \|f\|_{\mathcal{X}_0}.\end{aligned}$$

This shows that $\mathcal{X}_0$ continuously embeds into $\mathcal{X}_\alpha$.

Since $H_{d,1-\alpha} \circ H_{d,\alpha} = -S$, where $S(a)(\cdot) := a(\cdot + 1)$, and S is an isomorphism in $\mathrm{BMO}(\mathbb{Z})$, from the preceeding formula it follows that R_α is one-to-one and $R_\alpha^{-1} = -{R_0}^{-1} S^{-1} H_{d,1-\alpha}$. Then, the embedding is also onto.

The equivalence of the norms now follows from the open mapping theorem. □

Proof of Theorem 4. Let $f \in \mathrm{BMO}(e^{-i\pi z})$, that we can identify with $(\mathcal{B}^1_\pi)^*$. Then, the Hankel operator with symbol f, $H_f : \mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}$ is bounded. This in turn is equivalent to saying that the Toeplitz operator $T_f : \mathcal{B}^2_{\pi/2} \to \mathcal{B}^2_{\pi/2}$ is bounded. By [Car11] it follows that $\big((-1)^n f(n)\big) \in \mathrm{BMO}(\mathbb{Z})$. In order to show that f satisfies condition (Y), we use Theorem 2 and using (16) we write $f = P_\pi \varphi$, with $\varphi \in L^\infty$, so that if $R > 0$ is given, $z \in \mathbb{C}$ such that $|\operatorname{Re} z| \leqslant R$, then

$$|f(iy)| \leqslant \int_{I*} |\varphi(t)| \frac{2e^{\pi|y|}}{|y|}\, dt + C \int_{(I*)^c} |\varphi(t)| \frac{e^{\pi|y|}|y|}{|t|(|t|+|y|)}\, dt$$

Using the dominated convergence theorem one sees at once that f satisfies condition (Y). Therefore, $f \in \mathrm{BMO}(e^{-i\pi z})$ belongs to $\mathcal{X}_0$, and $\mathrm{BMO}(e^{-i\pi z})$ continuously embeds in $\mathcal{X}_0$. □

5. Compactness, VMO and preduality

Proof of Theorem 5. (1) Let $\eta \in C_0(\mathbb{R})$ then by (14) and (15) we have

$$\begin{aligned}P_\pi \eta &= e^{-i\pi(\cdot)} P_+ (e^{i\pi(\cdot)} \eta) - e^{i\pi(\cdot)} P_+ (e^{-i\pi(\cdot)} \eta) \\ &= e^{i\pi(\cdot)} P_- (e^{-i\pi(\cdot)} \eta) - e^{-i\pi(\cdot)} P_- (e^{i\pi(\cdot)} \eta).\end{aligned}$$

Since $P_\pm$ map $C_0(\mathbb{R})$ onto $\mathrm{VMOA}^\pm$, resp., we immediately see that $P_\pi\eta \in \mathrm{VMO}(e^{-i\pi z})$. In order to show that $P_\pi(C_0(\mathbb{R})) = \mathrm{VMO}(e^{-i\pi z})$, let $f \in \mathcal{Y}_0$. Then, $e^{\pm i\pi(\cdot)} f_0 \in \mathrm{VMOA}^\pm$, resp., so that there exist $\eta_\pm \in C_0(\mathbb{R})$ such that

$$e^{i\pi(\cdot)} f_0 = P_+\eta_+, \quad \text{and} \quad e^{-i\pi(\cdot)} f_0 = P_-\eta_-, . \tag{22}$$

Arguing as in the case of identity (17), we have that

$$f_0 = P_\pi(e^{-i\pi(\cdot)}\eta_+) = P_\pi(e^{i\pi(\cdot)}\eta_-);$$

hence $P_\pi(C_0(\mathbb{R})) = \mathrm{VMO}(e^{-i\pi z})$. The stated about the equivalence of norms follows now from Theorem 1.

(2) A function $h \in \mathcal{B}^1_\pi$ clearly defines an element L_h of $\mathrm{VMO}(e^{-i\pi z})^*$ with the same pairing between $\mathcal{B}^1_\pi$ and $\mathrm{BMO}(e^{-i\pi z})$, as in formula (8).

Next, let $L \in \mathcal{Y}_0^*$ and consider the bounded linear functional Φ on $C_0(\mathbb{R})$, $\Phi = L \circ P_\pi$. Then, by the Riesz representation theorem there exists a unique finite regular Borel measure μ on $\mathbb{R}$ that represents the functional, that is,

$$L(P_\pi\eta) = \int_{\mathbb{R}} \eta \, d\mu.$$

We wish to show that μ has a density, which coincides with the restriction of a function $\mathcal{B}^1_\pi$, that is, $\mu = h_0 dm$, with $h \in \mathcal{B}^1_\pi$. Observe that if $\mathrm{supp}(\widehat{\eta})$ does not intersect $[-\pi, \pi]$ so that $P_\pi\eta = 0$, then $\int_{\mathbb{R}} \eta \, d\mu = 0$. Hence, the distribution $\widehat{\mu}$ is supported on $[-\pi, \pi]$. By the classical Paley–Wiener–Schwartz theorem, there exists a function h in $\mathcal{E}_\pi$ such that $\mu = h_0$, as distributions. In particular, for all Schwartz functions $\widetilde{\eta}$, $\langle \widetilde{\eta}, \mu \rangle = \int_{\mathbb{R}} \widetilde{\eta} h_0 dm$. Approximating $\eta \in C_0(\mathbb{R})$ with Schwartz functions $\eta^{(\delta)}$ and passing to the limit we have

$$\int_{\mathbb{R}} \eta \, d\mu = \lim_{\delta \to 0} \int_{\mathbb{R}} \eta^{(\delta)} \, d\mu = \lim_{\delta \to 0} \int_{\mathbb{R}} \eta^{(\delta)}(x) h_0(x) \, dx = \int_{\mathbb{R}} \eta(x) h_0(x) \, dx.$$

Hence, $d\mu = h_0 dm$ and since μ has bounded total variation, $h_0 \in L^1(\mathbb{R})$, that is, $h \in \mathcal{B}^1_\pi$. Since $P_\pi h = h$, using part (1) and a simple approximation argument, we see that

$$L(\varphi) = L(P_\pi\eta) = \int_{\mathbb{R}} \eta(x) h_0(x) \, dx = \int_{\mathbb{R}} P_\pi\eta(x) h_0(x) \, dx,$$

that is, $L(\varphi) = L_h(\varphi)$, for all $\varphi \in \mathcal{Y}_0$, as we wished to show.

By part (1), there exists $C_1 > 0$ such that for every $\varphi \in \mathcal{Y}_0$, there exists $\eta_\varphi \in C_0(\mathbb{R})$ such that $P_\pi\eta_\varphi = \varphi$ and $\|\eta_\varphi\|_{L^\infty} \leqslant C_1 \|\varphi\|_{\mathrm{VMO}(e^{-i\pi z})}$. Then, for $h \in \mathcal{B}^1_\pi$,

$$\begin{aligned} \|L_h\|_{\mathrm{VMO}(e^{-i\pi z})^*} &= \sup_{\|\varphi\|_{\mathrm{VMO}(e^{-i\pi z})} \leqslant 1} |L_h(\varphi)| \\ &= \sup_{\|\varphi\|_{\mathrm{VMO}(e^{-i\pi z})} \leqslant 1} \Big| \int_{\mathbb{R}} h_0 \eta_\varphi \, dm \Big| \leqslant \sup_{\|\varphi\|_{\mathrm{VMO}(e^{-i\pi z})} \leqslant 1} C_1 \|h\|_{\mathcal{B}^1_\pi} \|\varphi\|_{\mathrm{VMO}(e^{-i\pi z})}, \end{aligned}$$

whence $\|L_h\|_{\mathrm{VMO}(e^{-i\pi z})^*} \leqslant C_1 \|h\|_{\mathcal{B}^1_\pi}$. On the other hand, if $h \in \mathcal{B}^1_\pi$,

$$\begin{aligned} \|h\|_{\mathcal{B}^1_\pi} &= \sup_{\eta \in C_0(\mathbb{R}),\, \|\eta\|_{L^\infty} \leqslant 1} \Big| \int_{\mathbb{R}} \eta(x) h_0(x) \, dx \Big| = \sup_{\eta \in C_0(\mathbb{R}),\, \|\eta\|_{L^\infty} \leqslant 1} |L_h(P_\pi\eta)| \\ &\leqslant \sup_{\eta \in C_0(\mathbb{R}),\, \|\eta\|_{L^\infty} \leqslant 1} \|L_h\|_{\mathrm{VMO}(e^{-i\pi z})^*} \|P_\pi\eta\|_{\mathrm{VMO}(e^{-i\pi z})} \leqslant C_2 \|L_h\|_{\mathrm{VMO}(e^{-i\pi z})^*}. \end{aligned}$$

This completes the proof. □

We point out that in the course of the proof, we used the following analouge of the theorem of F. and M. Riesz.

Proposition 5.1. *Let μ a finite Borel measure on $\mathbb{R}$ such that $\operatorname{supp}(\widehat{\mu}) \subseteq [-\pi, \pi]$. Then, there exists $h \in \mathcal{B}^1_\pi$ such that $d\mu = h_0 dm$. In particular, μ is absolutely continuous w.r.t. the Lebesgue measure.*

Proof of Theorem 6. Let $\varphi \in \mathcal{Y}_0$, $\varphi = P_\pi \eta$ with $\eta \in C_0(\mathbb{R})$, be given. Let $\{f_n\}$ be a sequence in $\mathcal{B}^2_{\pi/2}$ weakly converging to 0. It is well known, and easy to see, that this is equivalent to the conditions $\|f_n\|_{\mathcal{B}^2_{\pi/2}} \leqslant C$, with C independent of n and $f_n \to 0$ uniformly on compact subsets[7]. Given $\varepsilon > 0$, let $E \subseteq \mathbb{R}$ compact and n_ε positive integer be such that

$$\sup_{x \in E^c} |\eta(x)| < \varepsilon, \quad \text{and} \quad \sup_{x \in E, n \geqslant n_\varepsilon} |f_n(x)| < \varepsilon / |E|^{1/2}.$$

Notice that, since $f_n g \in \mathcal{B}^1_\pi$,

$$\begin{aligned} |\langle H_\varphi f_n \,|\, g \rangle| = |\langle \varphi \,|\, f_n g \rangle| = |\langle \eta \,|\, f_n g \rangle| &\leqslant \left(\int_E + \int_{E^c} \right) |\eta(x) f_n(x) g(x)| \, dx \\ &\leqslant \|\eta\|_{L^\infty} \varepsilon \|g\|_{\mathcal{B}^2_{\pi/2}} + C\varepsilon \|g\|_{\mathcal{B}^2_{\pi/2}} \\ &\leqslant C' \varepsilon \|g\|_{\mathcal{B}^2_{\pi/2}}. \end{aligned}$$

Then, for $n \geqslant n_\varepsilon$ we have

$$\|H_\varphi f_n\|_{\mathcal{B}^2_{\pi/2}} = \sup_{g \in \mathcal{B}^2_{\pi/2}, \|g\|_{\mathcal{B}^2_{\pi/2}} \leqslant 1} \left| \langle H_\varphi f_n \,|\, g \rangle \right| \leqslant C' \varepsilon.$$

Hence, (1) implies (2).

Next, let

$$\mathcal{H} := \left\{ \psi \in \mathcal{S}' : \, H_\psi \text{ is a compact Hankel operators on } \mathcal{B}^2_{\pi/2} \right\}.$$

The first part of the proof and the norm equivalence statement in Theorem 2 show that the mapping $J : \mathrm{VMO}(e^{-i\pi z}) \ni \varphi \mapsto H_\varphi \in \mathcal{H}$ is an isomorphism onto its image. Thus, $J(\mathrm{VMO}(e^{-i\pi z}))$ is a closed subspace of $\mathcal{H}$. If we show that $\mathcal{H}$ and $J(\mathrm{VMO}(e^{-i\pi z}))$ have the same dual, it would follow that they are actually equal. We use the identification of $\mathrm{VMO}(e^{-i\pi z})^*$ with $\mathcal{B}^1_\pi$.

If $f \in \mathcal{B}^1_\pi$, then define a linear functional on $\mathcal{H}$ by setting

$$\Lambda_f(H_\varphi) := \langle \varphi \,|\, f \rangle.$$

[7] We provide the details in the subsequent Lemma 5.2.

Note that Λ_f is well defined since, if $f = \sum_{j=1}^4 g_j h_j$, with $g_j, h_j \in \mathcal{B}^2_{\pi/2}$ and $\sum_{j=1}^4 \|g_j\|_{\mathcal{B}^2_{\pi/2}} \|h_j\|_{\mathcal{B}^2_{\pi/2}} \leqslant C_1 \|f\|_{\mathcal{B}^1_\pi}$, then

$$\begin{aligned} |\Lambda_f(H_\varphi)| &= |\sum_{j=1}^4 \langle \varphi \overline{g_j} \,|\, h_j\rangle| = |\sum_{j=1}^4 \langle H_\varphi g_j \,|\, h_j\rangle| \\ &\leqslant \|H_\varphi\|_{\mathcal{B}^2_{\pi/2}\to\mathcal{B}^2_{\pi/2}} \sum_{j=1}^4 \|g_j\|_{\mathcal{B}^2_{\pi/2}} \|h_j\|_{\mathcal{B}^2_{\pi/2}} \\ &\leqslant C_1 \|H_\varphi\|_{\mathcal{B}^2_{\pi/2}\to\mathcal{B}^2_{\pi/2}} \|f\|_{\mathcal{B}^1_\pi}. \end{aligned}$$

Therefore, $\|\Lambda_f\|_{\mathcal{H}^*} \leqslant C_1 \|f\|_{\mathcal{B}^1_\pi}$. Conversely, let $\Lambda \in \mathcal{H}^*$. By Hahn–Banach, we can extend Λ to a bounded linear functional $\widetilde{\Lambda}$ on the space of compact operators on $\mathcal{B}^2_{\pi/2}$, having the same norm. Hence, there exists a trace class operator $T := \sum_{k=1}^\infty \lambda_k \langle \cdot \,|\, u_k\rangle v_k$ on $\mathcal{B}^2_{\pi/2}$ that represents $\widetilde{\Lambda}$, where $\{u_k\}, \{v_k\}$ are orthonornal bases of $\mathcal{B}^2_{\pi/2}$, and $(\lambda_k) \in \ell^1$. Notice that $f := \sum_{k=1}^\infty \lambda_k u_k v_k \in \mathcal{B}^1_\pi$, and that

$$\begin{aligned} \Lambda(H_\varphi) = \widetilde{\Lambda}(H_\varphi) &:= \operatorname{tr}\left(T^* H_\varphi\right) = \sum_{j=1}^\infty \langle H_\varphi u_j \,|\, T u_j\rangle \\ &= \sum_{k=1}^\infty \lambda_j \langle H_\varphi u_j \,|\, v_j\rangle = \langle \varphi \,|\, f\rangle. \end{aligned}$$

Hence, $\Lambda = \Lambda_f$ and the norm equivalence follows from the open mapping theorem. □

In the previous theorem, we have used an equivalent condition which guarantees that the sequence $\{f_n\}$ weakly converges to 0. For the reader's convenience, we include the proof of this statement, taking $\kappa = \pi$ (instead of $\kappa = \pi/2$).

Lemma 5.2. *Let $\{f_n\}$ be a sequence in $\mathcal{B}^2_\pi$ Then, the following two conditions are equivalent:*

(i) *$\{f_n\}$ is weakly converging to 0;*
(ii) *there exists $C > 0$ independent of n such that $\|f_n\|_{\mathcal{B}^2_\pi} \leqslant C$, and $f_n \to 0$ uniformly on compact subsets.*

Proof. $(i) \Rightarrow (ii)$. The uniform boundedness principle gives that there exists $C > 0$ such that $\|f_n\|_{\mathcal{B}^2_\pi} \leqslant C$ for every $n \in \mathbb{N}$. For $j \in \mathbb{Z}$, let $K_j := \operatorname{sinc}(\cdot - j)$. Then $\{K_j\}$ is an orthonormal basis for $\mathcal{B}^2_\pi$ (see (2)). Fix a compact set $E \subseteq \mathbb{C}$ and $\varepsilon > 0$. Then, there exists j_E such that if $|j| > j_E$ and $z \in E$ we have

$$|K_j(z)| \leqslant \frac{C}{|j|} \quad \text{and} \quad C^2 \sum_{|j|>j_E} \frac{1}{j^2} < \varepsilon.$$

Then, for $z \in E$

$$\begin{aligned}|f_n(z)| &\leqslant \sum_{|j|\leqslant j_E} |f_n(j)\operatorname{sinc}(z-j)| + \sum_{|j|>j_E} |f_n(j)\operatorname{sinc}(z-j)| \\ &\leqslant \sum_{|j|\leqslant j_E} |f_n(j)| + \|f_n\|_{\mathcal{B}^2_\pi} \sum_{|j|>j_E} \frac{C^2}{j^2} \\ &\leqslant \sum_{|j|\leqslant j_E} |\langle f_n \mid \operatorname{sinc}(\cdot - j)\rangle| + \varepsilon.\end{aligned}$$

Now, since $\{f_n\}$ weakly converges to 0, we can select a positive integer n_ε large enough so that if $n \geqslant n_\varepsilon$

$$\sum_{|j|\leqslant j_E} |\langle f_n \mid \operatorname{sinc}(\cdot - j)\rangle| < \varepsilon.$$

This shows that $(i) \Rightarrow (ii)$.

Conversely, let $g \in \mathcal{B}^2_\pi$, $\varepsilon > 0$ be given. Let $E \subseteq \mathbb{R}$ be a compact set such that $\|\chi_{E^c} g_0\|_{L^2(\mathbb{R})} < \varepsilon$. Then,

$$\begin{aligned}|\langle f_n \mid g\rangle| &\leqslant \left(\int_E + \int_{E^c}\right) |f_n(x)g(x)|\,dx \\ &\leqslant \sup_{x\in E} |f_n(x)||E|^{1/2}\|g\|_{\mathcal{B}^2_\pi} + C\|\chi_{E^c} g_0\|_{L^2(\mathbb{R})}\end{aligned}$$

By selecting n large enough so that $\sup_{x\in E} |f_n(x)||E|^{1/2} < \varepsilon$, the conclusion follows. □

6. Final remarks

The Bernstein spaces are particular cases of de Branges spaces, see, for example, [dB68] and [Rom14]. Indeed the entire function $e^{-i\pi z}$ is a Hermite Biehler function and, if $1 \leqslant p \leqslant \infty$,

$$\mathcal{B}^p_\pi = \left\{ f \text{ entire } : f(z)/e^{-i\pi z},\ f^\#(z)/e^{-i\pi z} \in H^p(\mathbb{C}^+) \right\} = \mathcal{H}^p(e^{-i\pi\cdot}).$$

If E is any Hermite Biehler and $\alpha \in [0, 1)$, we observe that, when $z \in \mathbb{C}^+$,

$$\operatorname{Re}\left(\frac{e^{i\pi\alpha}E(z) + e^{-i\pi\alpha}E^\#(z)}{e^{i\pi\alpha}E(z) - e^{-i\pi\alpha}E^\#(z)}\right) > 0.$$

Therefore, thanks to the Herglotz theorem [dB68, Theorem 4], there exists a positive, Poisson integrable, Borel measure σ_α such that

$$\frac{e^{i\pi\alpha}E(z) + e^{-i\pi\alpha}E^\#(z)}{e^{i\pi\alpha}E(z) - e^{-i\pi\alpha}E^\#(z)} = ip_\alpha z + c_\alpha + \int_{\mathbb{R}} \left(\frac{1}{t-z} - \frac{t}{1+t^2}\right) d\sigma_\alpha(t).$$

The measures σ_α are the Clark measures associated to E. For a complete description of these measures, we refer to [CMR06, Chapter 9] and to [Sak07].

If $E(z) = e^{-i\pi z}$, we are able to compute explicitely the associated Clark measures. Indeed

$$\operatorname{supp}\sigma_\alpha = \left\{t \in \mathbb{R} : e^{-i\pi\alpha}e^{i\pi t} = e^{i\pi\alpha}e^{-i\pi t}\right\} = \left\{t \in \mathbb{R} : e^{2i\pi t} = e^{2i\pi\alpha}\right\} = \{t = n + \alpha : n \in \mathbb{Z}\} =: \mathbb{Z}_\alpha$$

and

$$\sigma_\alpha(n+\alpha) = \frac{1}{|(e^{-i\pi(\cdot)})'|} = \frac{1}{\pi}.$$

It follows that the space $\mathcal{X}_\alpha$ can be equivalently described through the Clark measures.

Corollary 6.1. *Let $\alpha \in [0,1)$. Then*

$$\mathcal{X}_\alpha = \Big\{ f \in \mathcal{E}_\pi : f(iy) = o(|y|e^{\pi|y|}) \text{ for } |y| \to \infty,\ f(\cdot)/e^{-i\pi(\cdot)} \in \mathrm{BMO}(\sigma_\alpha) \Big\} \Big/ \operatorname{span}\{e^{\pm i\pi(\cdot)}\},$$

with norm

$$\|f\|_{\mathcal{X}_\alpha} = \pi \|f(\cdot)/e^{-i\pi(\cdot)}\|_{\mathrm{BMO}(\sigma_\alpha)}.$$

This last description of the space $\mathcal{X}_\alpha$ is similar to the one presented in [Bes15, Theorem 2].

Some of the results presented in this work, hold also while describing the dual of $\mathcal{H}^1(E)$ when $E(z)$ is a general Hermite-Biehler function. We will present these theorems in a different article. Nevertheless, the Bernstein space $\mathcal{B}^1_\pi$ satisfies some exceptional properties: because $H^1(\mathbb{Z})$ is isomorphic to $\mathcal{B}^1_\pi$, we are able to provide an atomic description for $\mathcal{B}^1_\pi$. We first recall the definition of the atoms of $H^1(\mathbb{Z})$.

Definition 6.2. A sequence $\alpha = (\alpha_n)$ is an atom of $H^1(\mathbb{Z})$ if it satisfies these three conditions:

- α has compact (i.e., finite) support, contained in an interval A;
- $\sup_n |\alpha_n| \leqslant 1/\#A$, where $\#A$ denotes the cardinality of A;
- α has mean value 0, that is, $\sum_{n\in I} \alpha_n = 0$.

Definition 6.3. The atoms a of the space $\mathcal{B}^1_\pi$ are the functions

$$a(z) := \sum_n (-1)^n \alpha_n \ \operatorname{sinc}(\pi(z-n)) \ , \tag{23}$$

where (α_n) is an atom of $H^1(\mathbb{Z})$.

If $f \in \mathcal{B}^1_\pi$, the sequence (f_n), where $f_n := (-1)^n f(n)$ belongs to $H^1(\mathbb{Z})$. Consequently, for every $\varepsilon > 0$, there exists a family of atoms $\{\alpha^j\}$, $\alpha^j = (\alpha^j_n)$ such that

$$\|(f_n) - \sum_j \lambda_j(\alpha^j)\|_{H^1(\mathbb{Z})} < \varepsilon.$$

From [Eof95], we know that $\widetilde{T}_0 : H^1(\mathbb{Z}) \to \mathcal{B}^1_\pi$ is an isomorphism. It follows that

$$\Big\| f - \sum_j \lambda_j \Big(\sum_n (-1)^n \alpha^j_n \ \operatorname{sinc}(\pi(\cdot - n)) \Big) \Big\|_{\mathcal{B}^1_\pi} \leqslant C \|(f_n) - \sum_j \lambda_j(\alpha^j)\|_{H^1(\mathbb{Z})} < C\varepsilon.$$

Another interstenting properties that we will study in the future, is the atomic decomposition of the de Branges space $\mathcal{H}^1(E)$. Indeed it will be interesteng to characterize for which Hermite-Biehler function, the space $\mathcal{H}^1(E)$ admits an atomic decomposition. We think that the one-component Hermite-Biehler functions, used for example in [Bar05, Bes15], will play a fundamental role.

References

[And14] N. B. Andersen, *Entire L^p-functions of exponential type*, Expo. Math. **32** (2014), no. 3, 199–220. MR 3253566

[Bar05] Anton Baranov, *Bernstein-type inequalities for shift-coinvariant subspaces and their applications to carleson embeddings*, Journal of Functional Analysis - J FUNCT ANAL **223** (2005), 116–146.

[BBK11] Anton Baranov, Roman Bessonov, and Vladimir Kapustin, *Symbols of truncated Toeplitz operators*, J. Funct. Anal. **261** (2011), no. 12, 3437–3456. MR 2838030

[BC98] Santiago Boza and Maria J. Carro, *Discrete Hardy spaces*, Studia Math. **129** (1998), no. 1, 31–50. MR 1611926

[Ber23] S. N. Bernstein, *Sur une propriété des fonctions entières*, C. R. Acad.Sci. **176** (1923), 1603–1605.

[Bes15] R. V. Bessonov, *Duality theorems for coinvariant subspaces of H^1*, Adv. Math. **271** (2015), 62–90. MR 3291857

[Car11] Marcus Carlsson, *On truncated Wiener-Hopf operators and* BMO($\mathbb{Z}$), Proc. Amer. Math. Soc. **139** (2011), no. 5, 1717–1733. MR 2763760

[CMR06] Joseph. A. Cima, Alec Matheson, and William T. Ross, *The Cauchy transform*, Mathematical surveys and Monographs, vol. 125, American Matheatical Society, 2006.

[CP22a] Mattia Calzi and Marco M. Peloso, *Bernstein spaces on Siegel CR manifolds*, Anal. Math. Phys. **12** (2022), no. 5, Paper No. 123, 33. MR 4483399

[CP22b] Mattia Calzi and Marco M. Peloso, *Carleson and sampling measures on bernstein spaces on siegel cr manifolds*, 2022.

[CW77] Ronald R. Coifman and Guido Weiss, *Extensions of Hardy spaces and their use in analysis*, Bull. Amer. Math. Soc. **83** (1977), no. 4, 569–645. MR 447954

[dB68] L. de Branges, *Hilbert spaces of entire functions*, Prentice-Hall, Inc., Englewood Cliffs, N.J., 1968. MR 0229011

[Eof95] Carolyn Eoff, *The discrete nature of the Paley-Wiener spaces*, Proc. Amer. Math. Soc. **123** (1995), no. 2, 505–512. MR 1219724

[Gar07] John B. Garnett, *Bounded analytic functions*, first ed., Graduate Texts in Mathematics, vol. 236, Springer, New York, 2007. MR 2261424

[Lan99] Serge Lang, *Complex analysis*, fourth ed., Graduate Texts in Mathematics, vol. 103, Springer-Verlag, New York, 1999. MR 1659317

[Lev96] B. Ya. Levin, *Lectures on entire functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, Providence, RI, 1996, In collaboration with and with a preface by Yu. Lyubarskii, M. Sodin and V. Tkachenko, Translated from the Russian manuscript by Tkachenko. MR 1400006

[MPS21] Alessandro Monguzzi, Marco M. Peloso, and Maura Salvatori, *Fractional Paley-Wiener and Bernstein spaces*, Collect. Math. **72** (2021), no. 3, 615–643. MR 4297147

[MPS22] ———, *Sampling in spaces of entire functions of exponential type in $\mathbb{C}^{n+1}$*, J. Funct. Anal. **282** (2022), no. 6, Paper No. 109377, 33. MR 4362860

[Nik02] Nikolai K. Nikolski, *Operators, functions, and systems: an easy reading. Vol. 1*, Mathematical Surveys and Monographs, vol. 92, American Mathematical Society, Providence, RI, 2002, Hardy, Hankel, and Toeplitz, Translated from the French by Andreas Hartmann. MR 1864396

[O'L22] Ryan O'Loughlin, *Symbols of compact truncated toeplitz operators*, Journal of Mathematical Analysis and Applications **507** (2022), no. 2, 125819.

[PP37] M. Plancherel and G. Pólya, *Fonctions entières et intégrales de fourier multiples*, Comment. Math. Helv. **10** (1937), no. 1, 110–163. MR 1509570

[PW34] R. E. A. C. Paley and N. Wiener, *Fourier transforms in the complex domain*, American Mathematical Society Colloquium Publications, vol. 19, American Mathematical Society, Providence, RI, 1987[1934], Reprint of the 1934 original. MR 1451142

[Roc87] Richard Rochberg, *Toeplitz and Hankel operators on the Paley-Wiener space*, Integral Equations Operator Theory **10** (1987), no. 2, 187–235. MR 878246

[Rom14] R. Romanov, *Canonical systems and de Branges spaces*, arXiv e-prints (2014), arXiv:1408.6022.

[Sak07] Eero Saksman, *An elementary introduction to clark measures*, 85–136 (English).

[Tri10] Hans Triebel, *Theory of function spaces*, Modern Birkhäuser Classics, Birkhäuser/Springer Basel AG, Basel, 2010, Reprint of 1983 edition [MR0730762]. MR 3024598

Dipartimento di Matematica "F. Enriques", Università degli Studi di Milano, Via C. Saldini 50, I-20133 Milano

Email address: carlo.bellavita@gmail.com

Dipartimento di Matematica "F. Enriques", Università degli Studi di Milano, Via C. Saldini 50, I-20133 Milano

Email address: marco.peloso@unimi.it